ITAO7106 Marketing Analytics

Semester 2 Assignment

Market STP strategy and 4P marketing mix

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1. Introduction and Background

The grocery retailing industry is highly competitive, and private label products are a key area of differentiation and margin enhancement (Ailawadi and Harlam, 2004). Retailers are no longer able to use category-level averages to inform new product decisions as consumer preferences become more and more diverse. Rather, they need to determine which customer segments are most valuable, what product attributes are important to these segments, and how to use analytical insights to create a marketing mix that will lead to commercial action. The fact that there is a lot of heterogeneity in consumer needs is one of the reasons why market segmentation is an important concept in modern marketing, as argued by Wedel and Kamakura (2000), who state that if a firm does not consider this heterogeneity, it will misallocate its resources and will not be able to satisfy any segment.

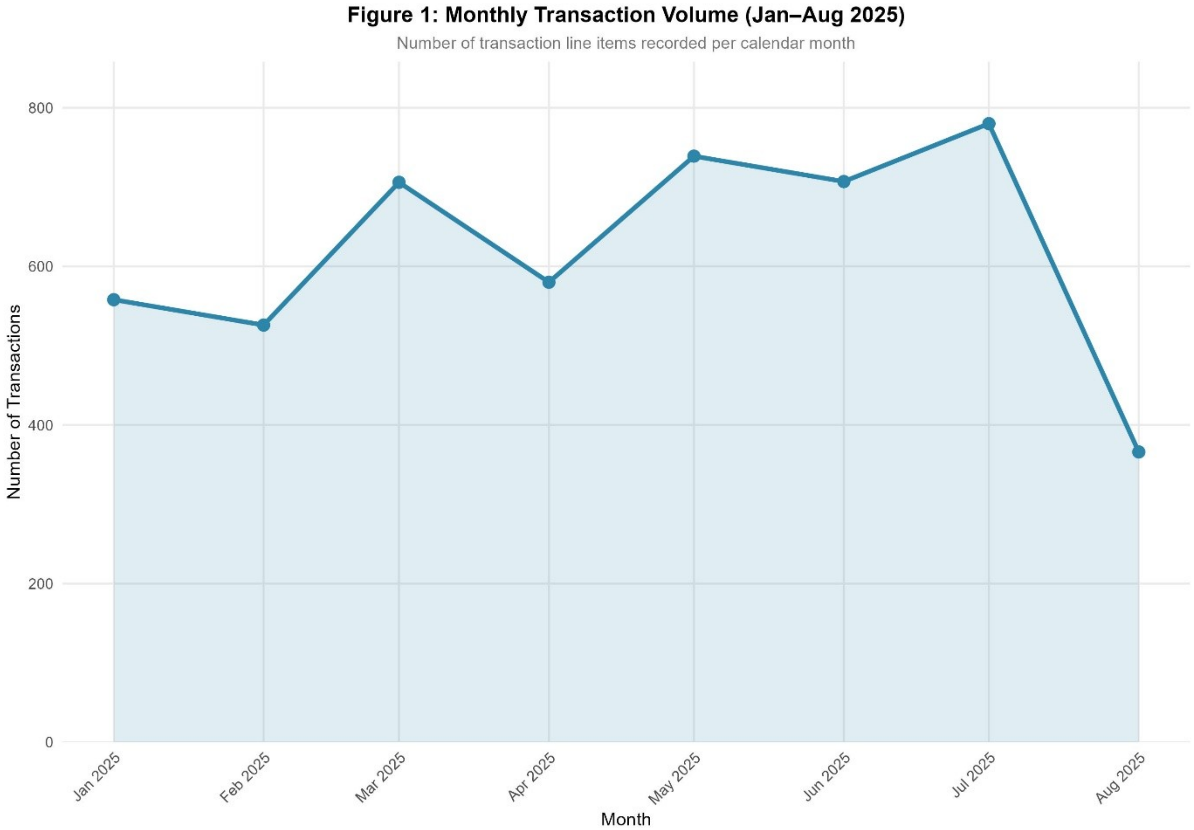
This report tackles just that issue. A UK grocery store with 922 existing customers and 5,000 potential customers wants to introduce a new private label coffee product. Coffee is one of the most competitive categories in grocery retail, and consumers are split on their price sensitivity, format, roast strength, origin, and sustainability. The wrong target segment or a poor product design may make the launch unprofitable.

To overcome this, the report uses four analytical techniques: exploratory data analysis, RFM-based cluster analysis, conjoint analysis and principal component analysis to develop an evidence-based segmentation, targeting and positioning strategy, which finally leads to a recommended 4P marketing mix for the new product.

2. Exploratory Data Analysis

2.1 Methodology

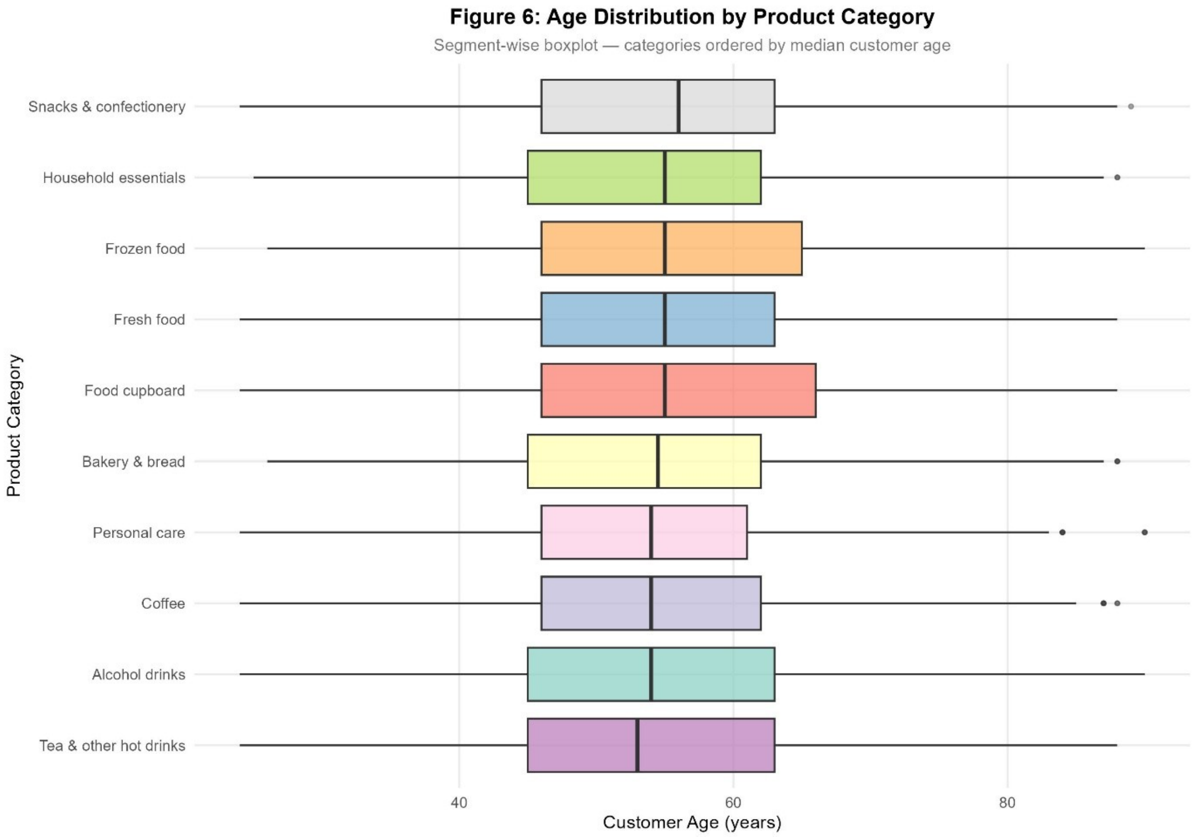
To perform any segmentation or predictive modelling, it was necessary to understand the structure, quality and distributional characteristics of the data. The first step in any customer analytics pipeline is exploratory data analysis (EDA), which allows the analyst to uncover anomalies, determine the distribution of variables, and identify patterns that will help guide modelling decisions (Hair et al., 2019). In this stage, two datasets were used: retail transaction dataset with 5,000 rows and 14 variables related to purchase behaviour and customer demographics, and prospect customer dataset with 5,000 rows and 8 variables related to customer demographics.



The retail data was cleaned in a step-by-step manner. First, two rows were deleted, both of which were fully duplicated, which would otherwise artificially boost spending numbers. Second, 36 rows with a ProductID of 'PNA' (not available) were removed because they did not have a valid product identifier and would have no contribution to the category level analysis. Third, rows where there was no ProductCategory value were dropped, resulting in a final working data set of 4,962 transaction records. The UnitPrice variable was then multiplied by the Quantity variable to create a TotalSpend variable, which gave a monetary value to each transaction line item.

One of the key methodological choices made at this point was the handling of extreme values. A Winsorisation method was used with a 3×IQR fence, instead of deleting outlier rows, which would result in the loss of otherwise valid customer records. This method does not eliminate extreme values, but rather sets them to the extremes of what is statistically plausible, thereby maintaining the sample size without causing distortion (Tukey, 1977). Both UnitPrice and Quantity were recorded at a maximum of £1,592 and a mean of £4.13, and a maximum of 1,930 and a mean of about 10 for Quantity, respectively, which are both highly improbable for a single grocery item. UnitPrice was capped at £12.77 and Quantity at 45 after the capping. The capped values were then used to recalculate TotalSpend. All categorical demographic variables (Married, Education, Work) were labelled factor variables for

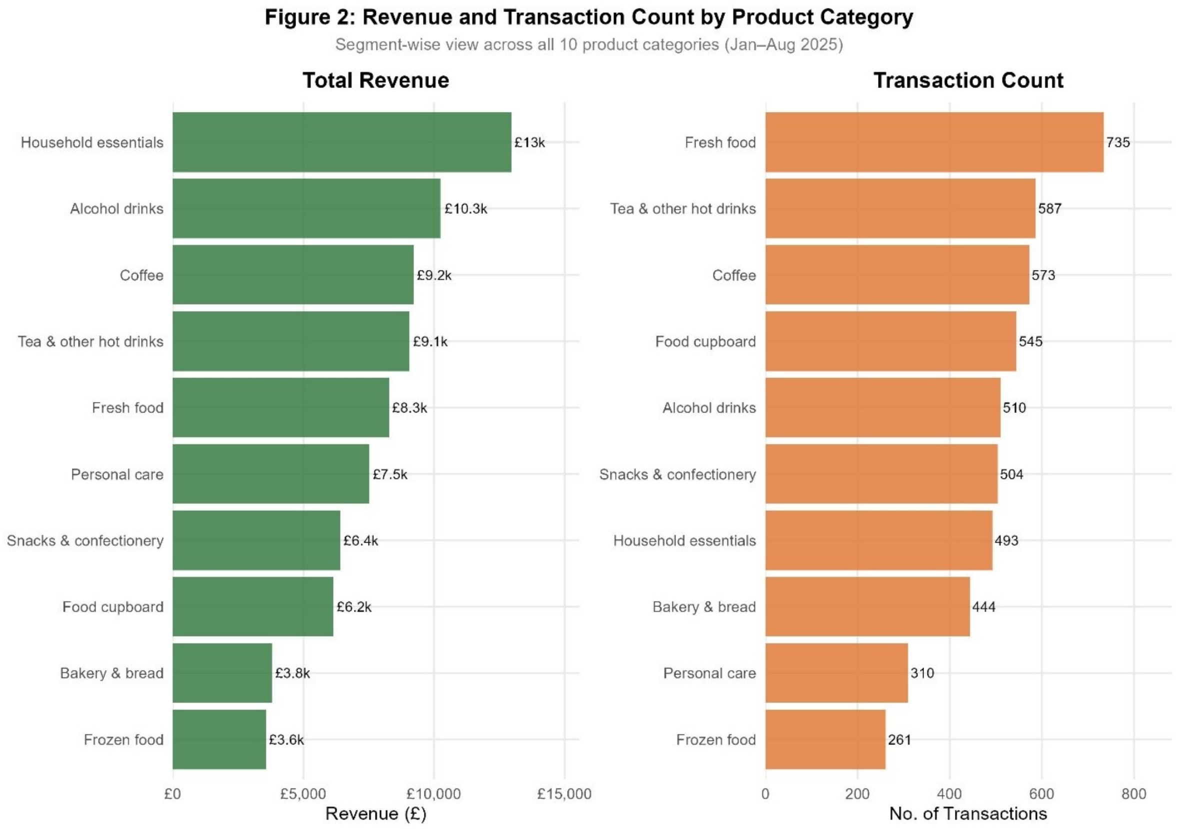
ease of visualisation and downstream modelling. The prospect dataset did not need any row level cleaning because there were no missing values or any duplicate rows.



2.2 Results and Discussion

The cleaned retail data showed some important patterns for commerce. The 4,962 transactions were made by 922 distinct customers, with 3,093 distinct invoices, indicating that many customers made multiple transactions during the observation period (January to August 2025). The volume of transactions was broadly stable over the month, with no significant seasonal variations, which is important as it suggests that the data is not a promotional blip or a seasonal effect, but a reflection of normal purchasing habits.

Coffee was a moderate performer in both revenue and transaction frequency, across the ten product categories, suggesting a presence of the category but not a dominant one.



This is commercially significant: a private label product that is entering a category where there is existing demand but not saturated demand is more likely to gain incremental share than a product that is entering a market that is too early or already well defended by branded products.

Demographic analysis of the 922 unique customers showed a relatively mature shopper base with a mean age of around 55 years, a mean household income of £68,900 and a mean household size of 2.73.

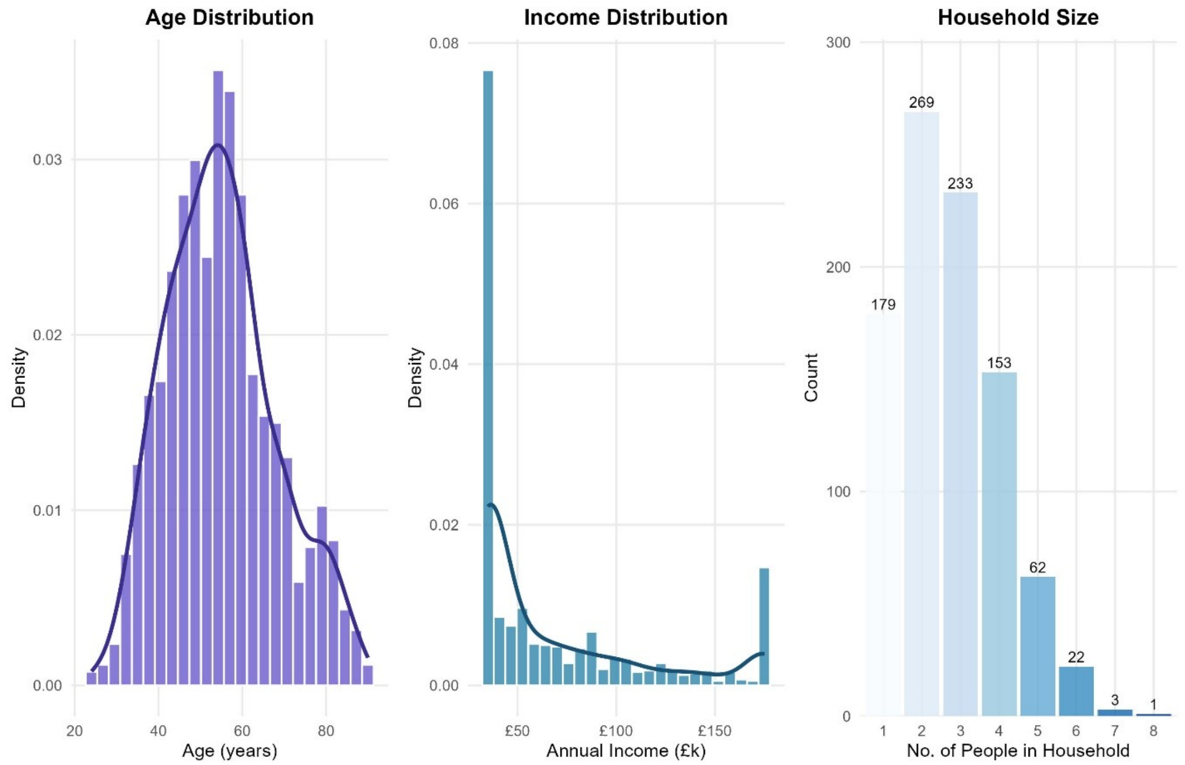


Figure 3: Customer Demographics — Age, Income & Household Size

One record per unique current customer (n = 925)

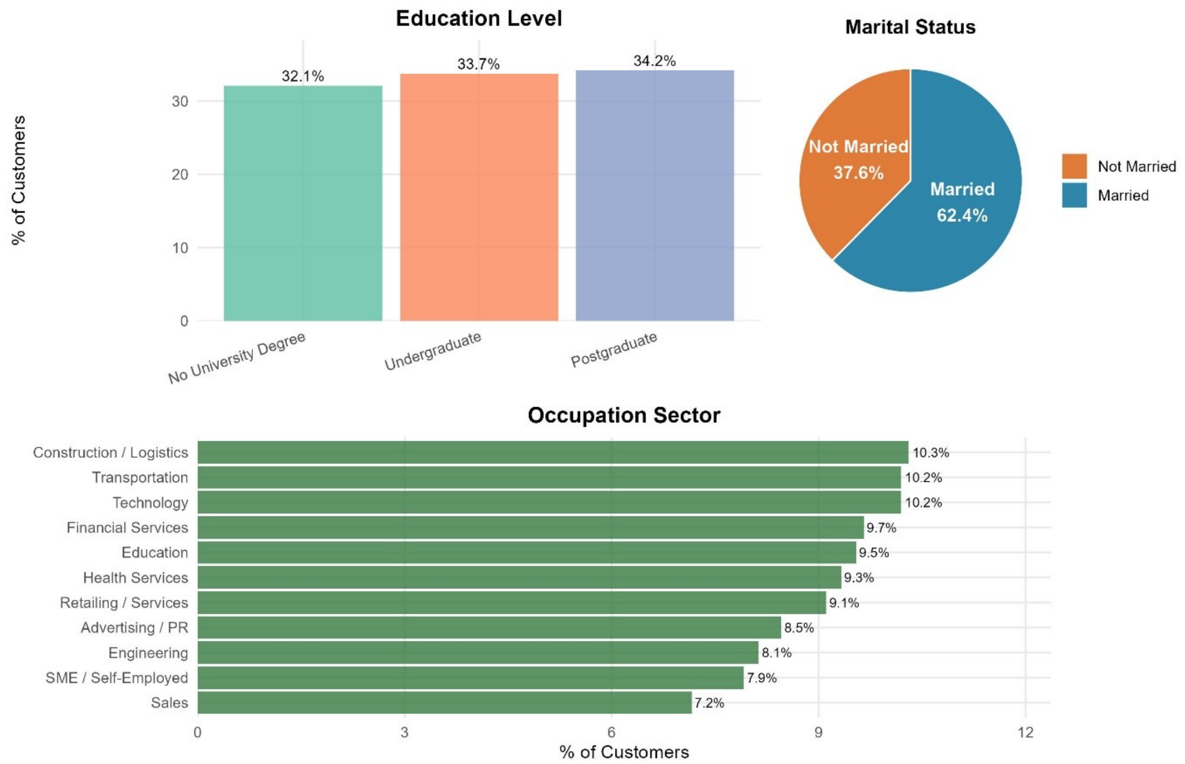
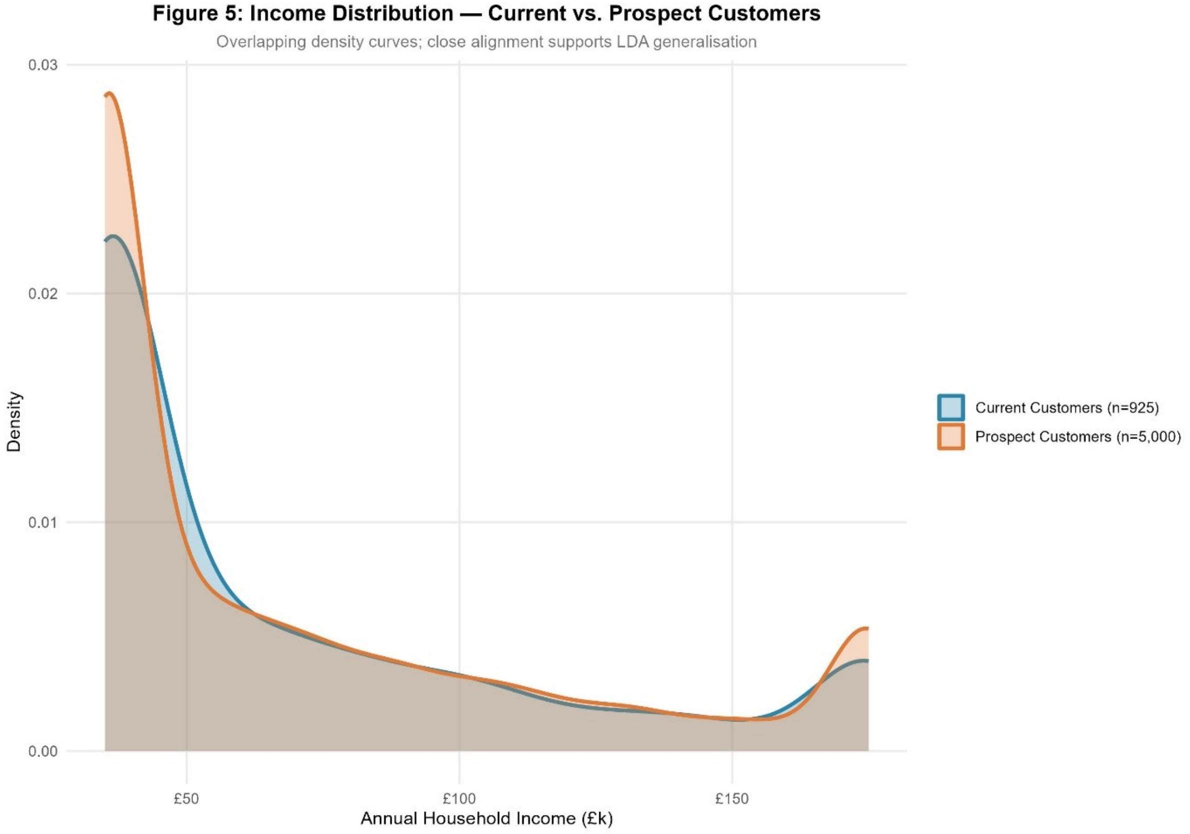


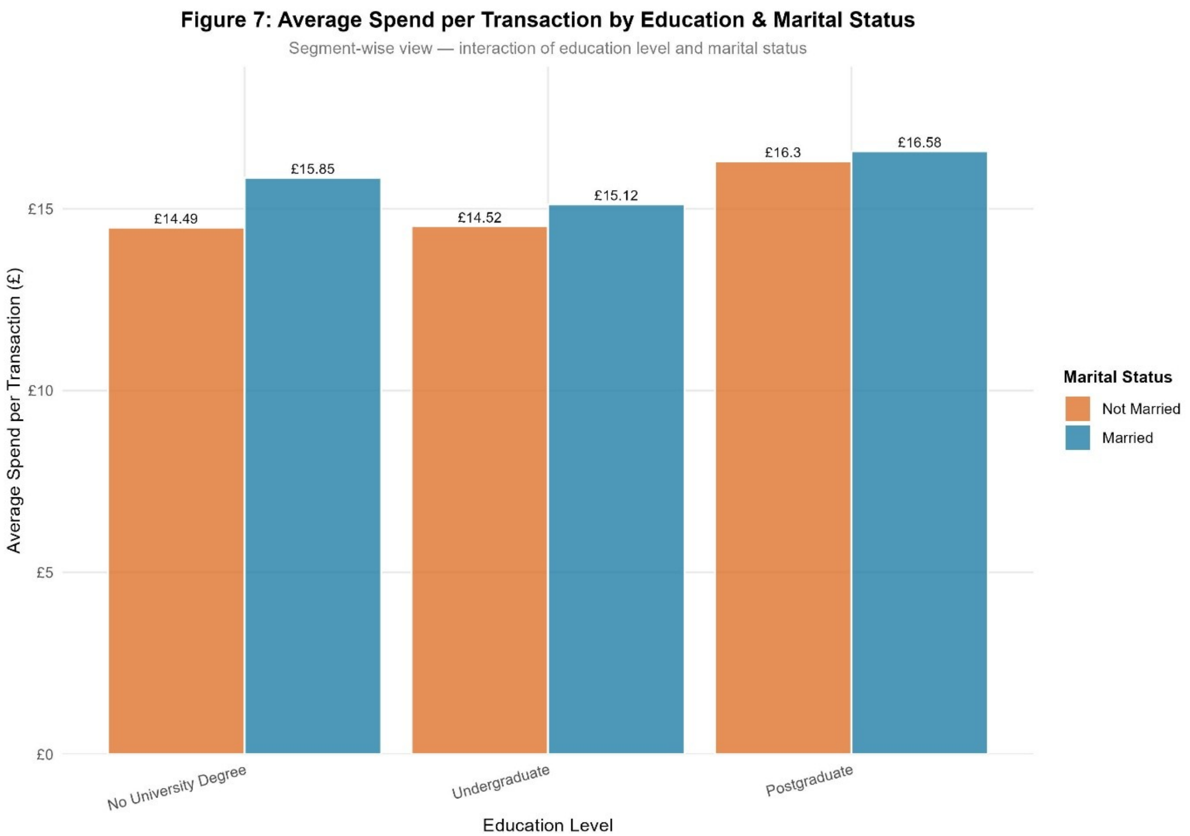
Figure 4: Education, Marital Status & Occupation of Current Customers

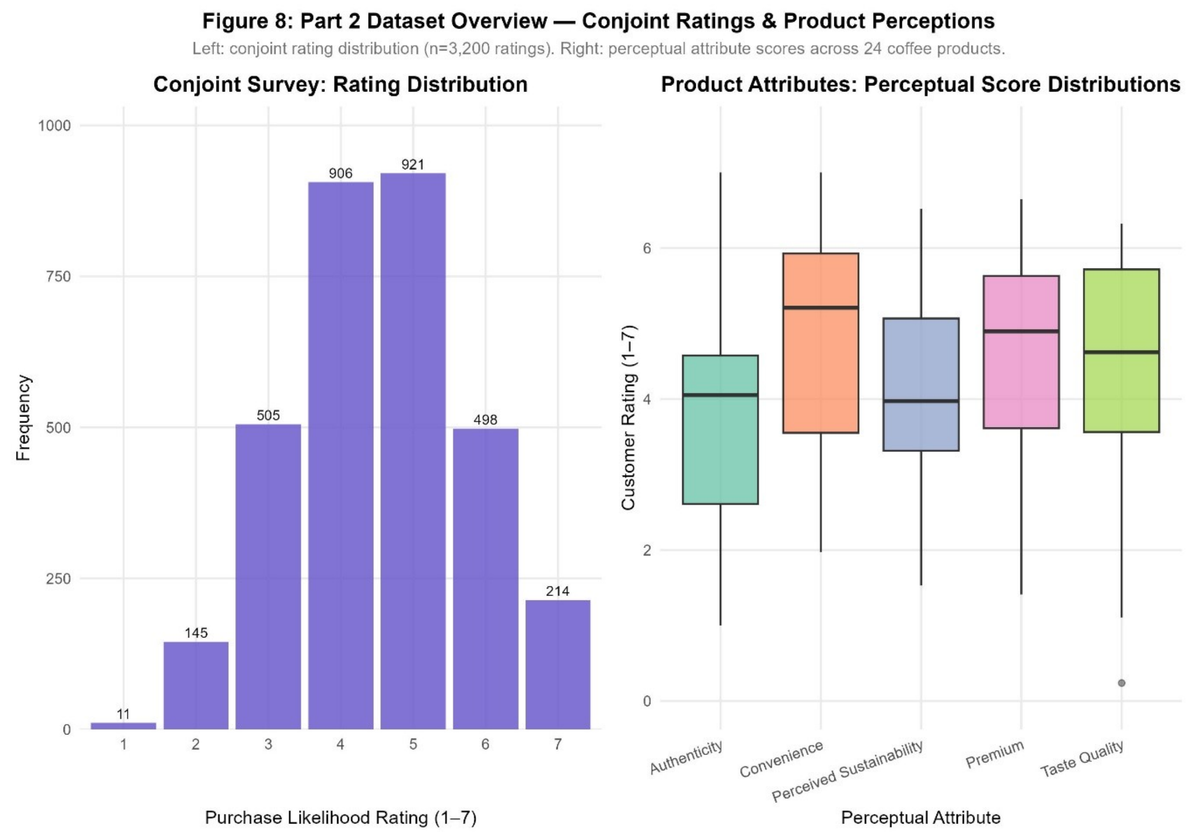
Segment-wise demographic breakdown (n = 925 unique customers)

The education profile was fairly flat across the three levels (no university degree, undergraduate, postgraduate), indicating that the retailer's customer base is not limited to any particular socioeconomic level. Most

importantly, the income and age distribution of the current customers was similar to that of the 5,000 prospect customers, a necessary condition for the validity of the Linear Discriminant Analysis used to classify prospects into behavioural segments in Section 3.





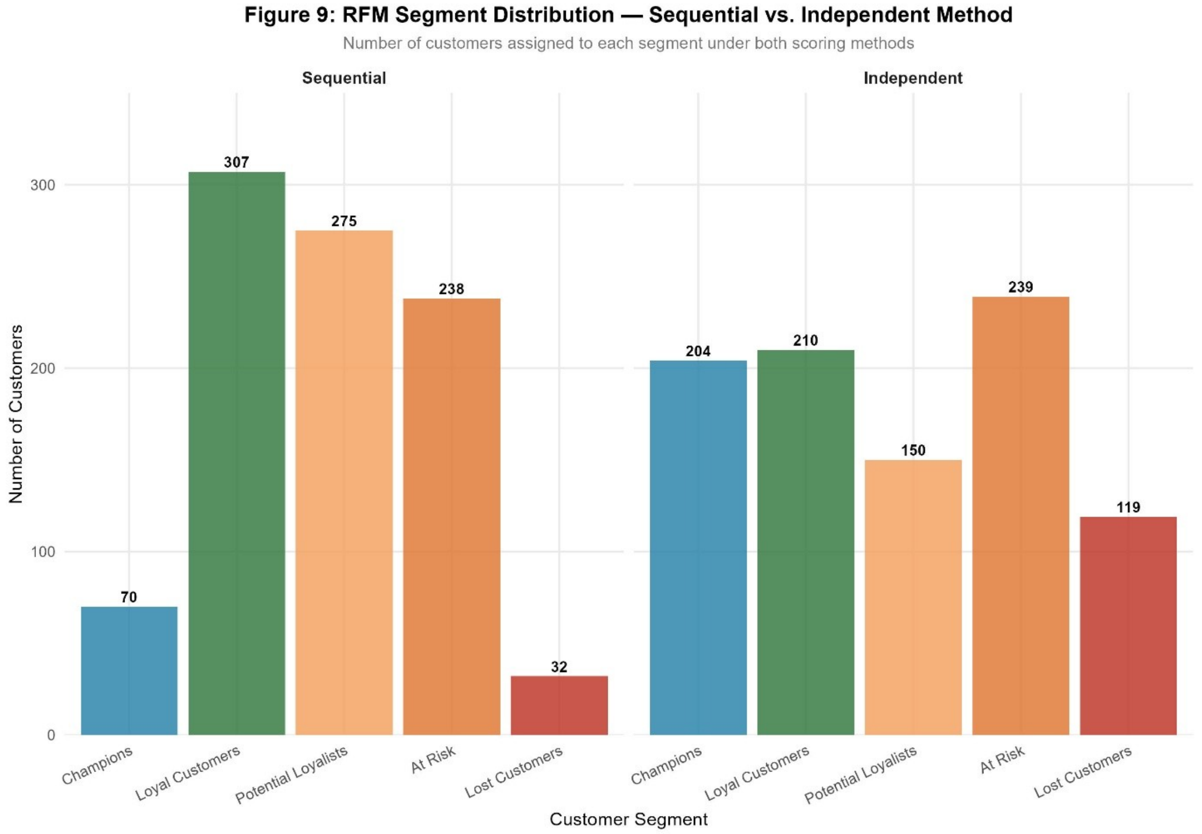


3. Cluster Analysis

3.1 Methodology

After determining the distributional properties of the data using EDA, the next step was to determine naturally occurring groups of customers based on their purchasing behaviour. The purpose of cluster analysis is to group customers into clusters so that customers in each cluster are as similar as possible to one another, and as different as possible from customers in other clusters (Everitt et al., 2011). Most importantly, this analysis is based on behavioural data from real transactions, not on assumptions about the demographics. This is important because two customers with the same age, income and household size can have very different behaviours as consumers, and it is behaviour, not background, that should be the basis for targeting.

The behavioural framework adopted was Recency, Frequency and Monetary value (RFM analysis). Recency is the number of days since the last purchase; Frequency is the number of different transactions; Monetary is the total amount spent during the observation period. These three dimensions together reflect the most commercially relevant aspects of a customer's relationship with the retailer (Hughes, 1994). The Recency was calculated using a reference date of 17 August 2025, one day after the last transaction in the data set, so that the most recently active customer would have the lowest score. The results showed that the two methods of computing RFM scores (sequential and independent quintile) agreed on the segment assignment for only 48.3% of the customers, indicating that the choice of scoring method is not a trivial one and can have a significant impact on who is targeted.



Before clustering, all three RFM dimensions were standardised to zero mean and unit variance. If the numbers were not standardised, the bigger numbers of the Monetary dimension would overwhelm the distance calculations and the clusters would be based on spend differences only, which would be misleading. First, a hierarchical clustering with Ward's minimum variance linkage was performed. Ward's method combines customers at each stage in a manner that reduces the overall variance within clusters, resulting in small, distinct clusters. The resulting dendrogram clearly indicated four clusters, the largest gap in merge heights being at the point where the dendrogram was divided into four clusters. This was confirmed formally by the elbow method, which shows that the total within cluster sum of squares decreased significantly from k=3 to k=4 and then decreased at a lesser rate. K-means clustering was then applied with k=4 to three different random starting configurations (seeds 42, 123, and 2025), each yielding the same total WSS of 727.68, which shows very high solution stability and that the solution is not a random result of the initial placement of centroids.

Figure 11: Dendrogram — Hierarchical Clustering (Ward's Method)

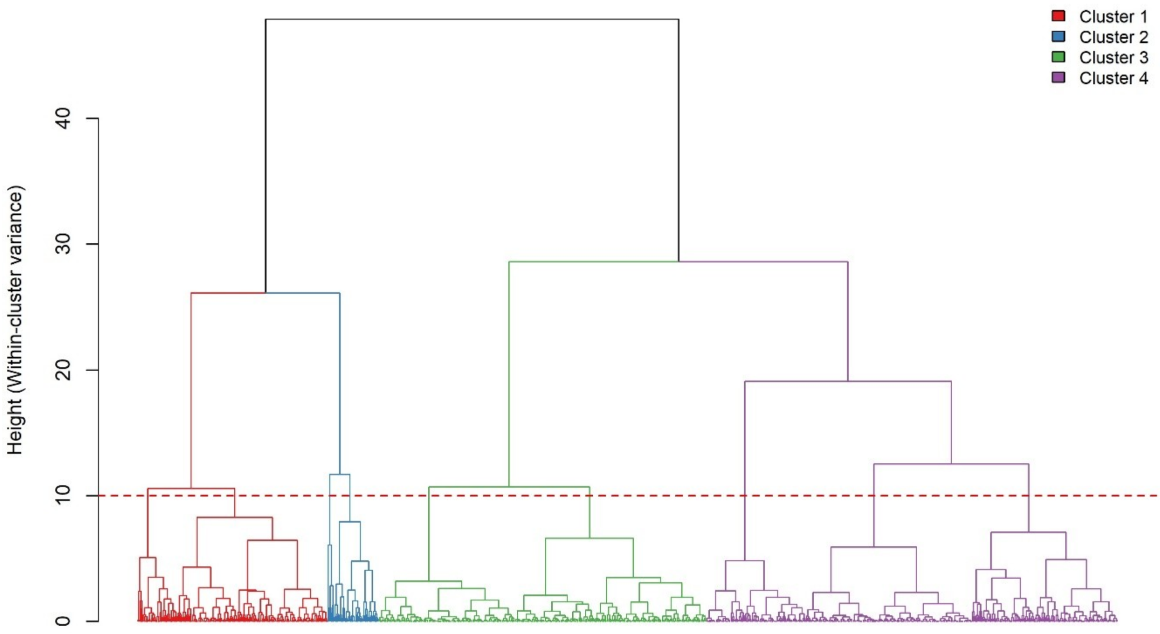
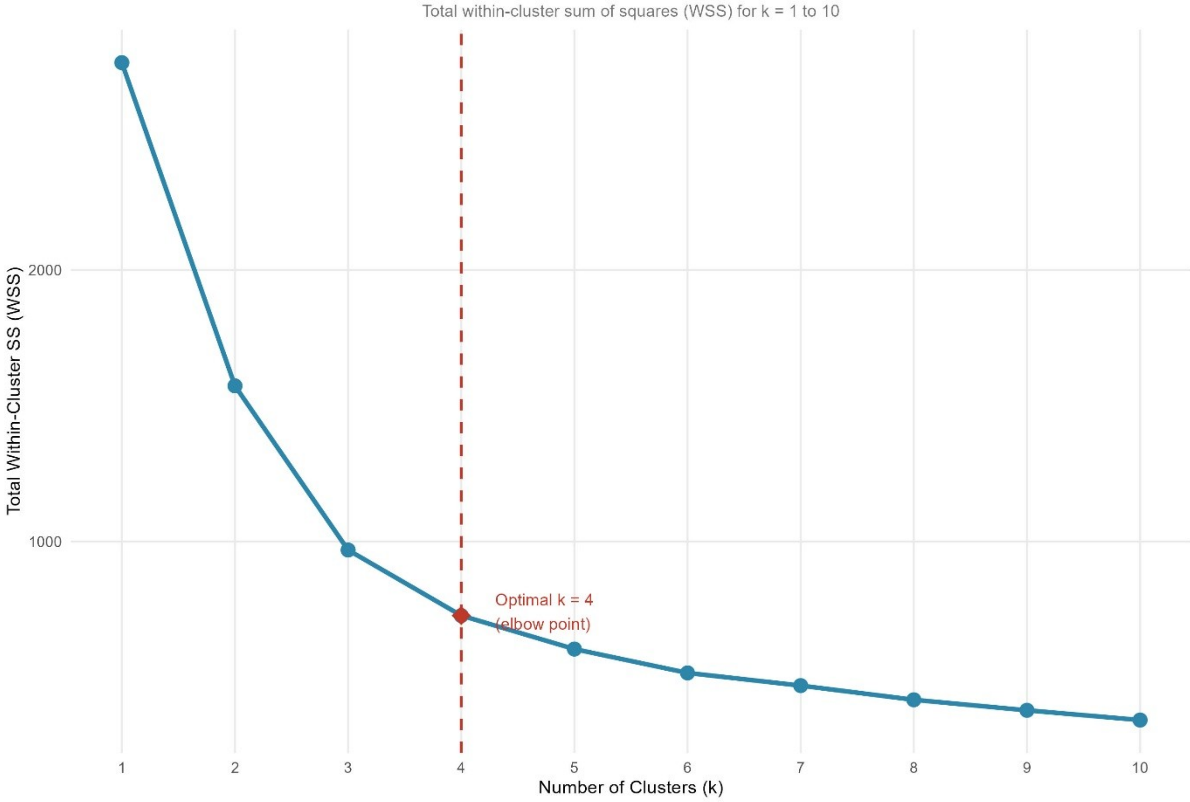
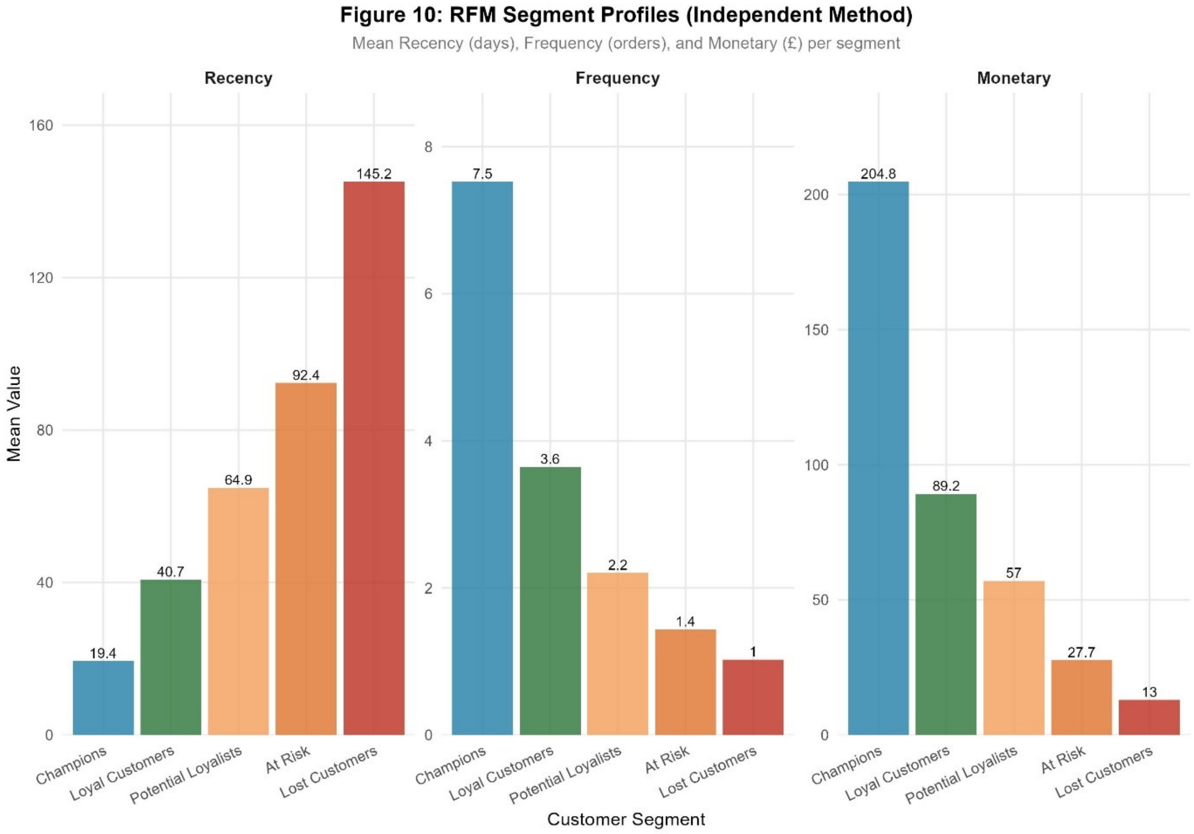


Figure 12: Elbow Method — Optimal Number of K-Means Clusters

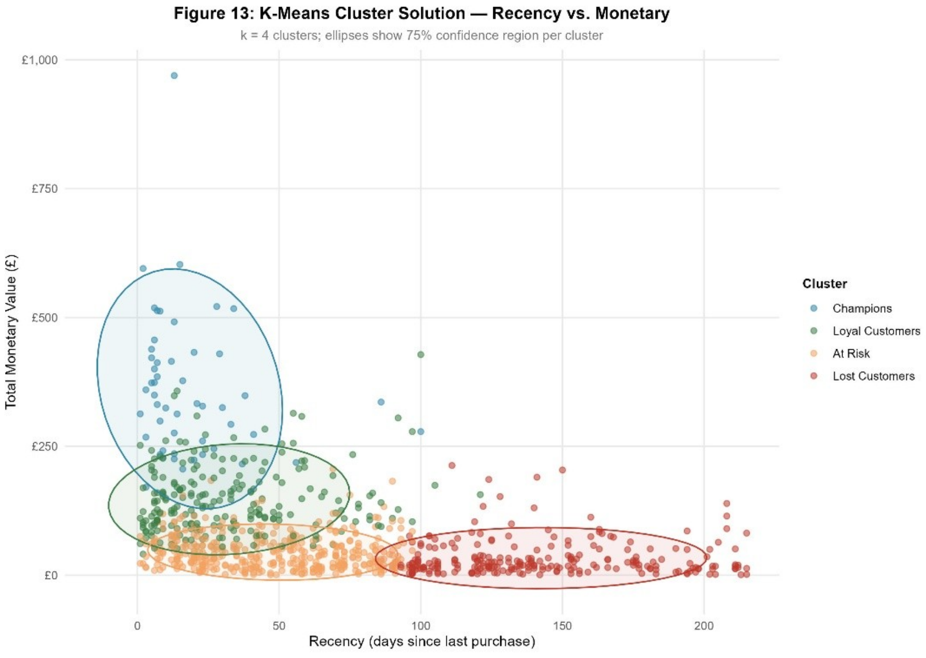


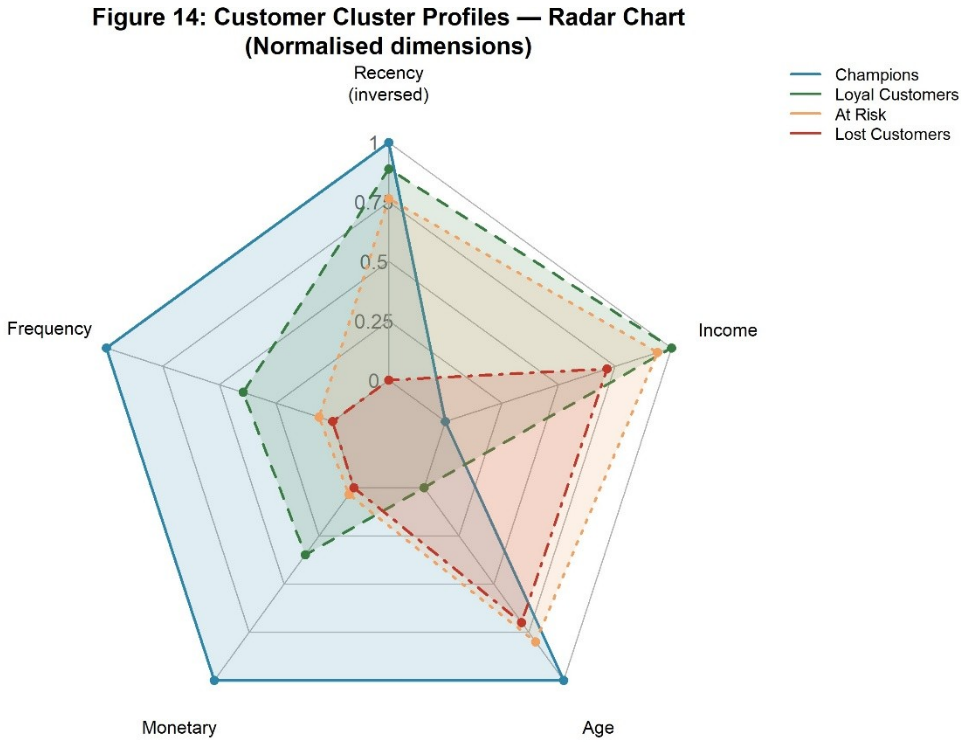
3.2 Segmentation Results



Segment 1: Champions (n=51)

The retailer's most valuable customers are the Champions. They have a low Recency (18.5 days), a low Frequency (12.2 orders) and a high Monetary value (£362). But their size is small, only 5.5% of the customers, so they are not a good target for a new product introduction. Investing in a group that is already fully engaged can be a replacement of spending, not an increase in revenue.





Segment 2: Loyal Customers (n=220)

Loyal Customers are the strategic sweet spot of the customer base. Their average Recency of 32.4 days indicates that they are actively engaged with the retailer, their Frequency of 5.72 orders suggests repeat purchases, not one-off transactions, and their average Monetary value of £147 puts them in the mid to high spending bracket. They are a mature, well-off population, with an average age of 53.8 years, a mean household income of £68,900 and a household size of 2.84, with 60.9% married. The distribution of education is fairly even at all three levels, indicating that this segment is not confined to any particular socioeconomic stratum.

Segment 3: At Risk (n=414)

The largest segment of At-Risk customers is 44.9% of the base. The average Recency of 47.8 days, Frequency of 2.1 orders and Monetary value of £44.80 indicate customers who have bought in the past but are not engaged. Interestingly, their age and income distribution is very similar to that of Loyal Customers, indicating that the differences between these segments are purely behavioural and not demographic, and therefore having direct implications for the design of re-engagement strategies.

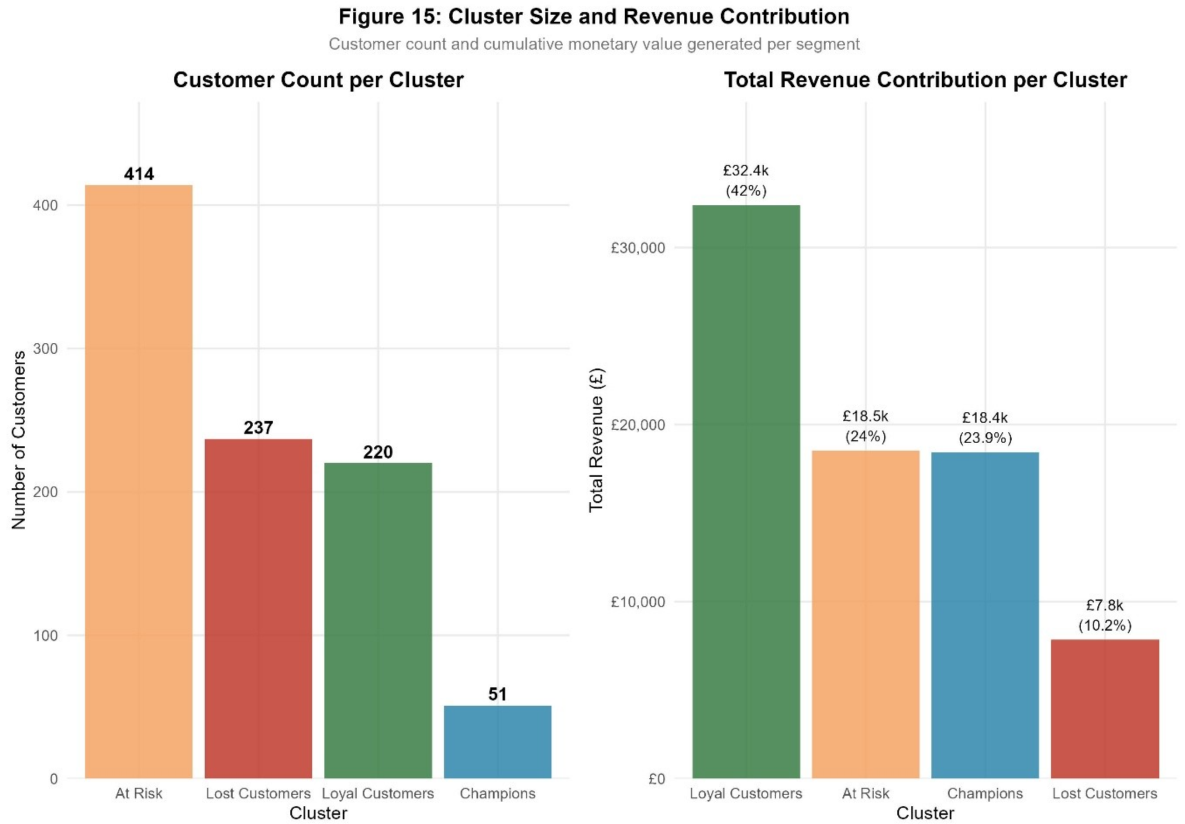
Segment 4: Lost Customers (n=237)

Lost Customers have not ordered an average of 142 days, with a Frequency of 1.45 orders and a Monetary value of £33.10. These

customers have essentially walked away from the retailer. Reactivation would be a marketing investment with an uncertain return, and therefore would not be a suitable target for any new product.

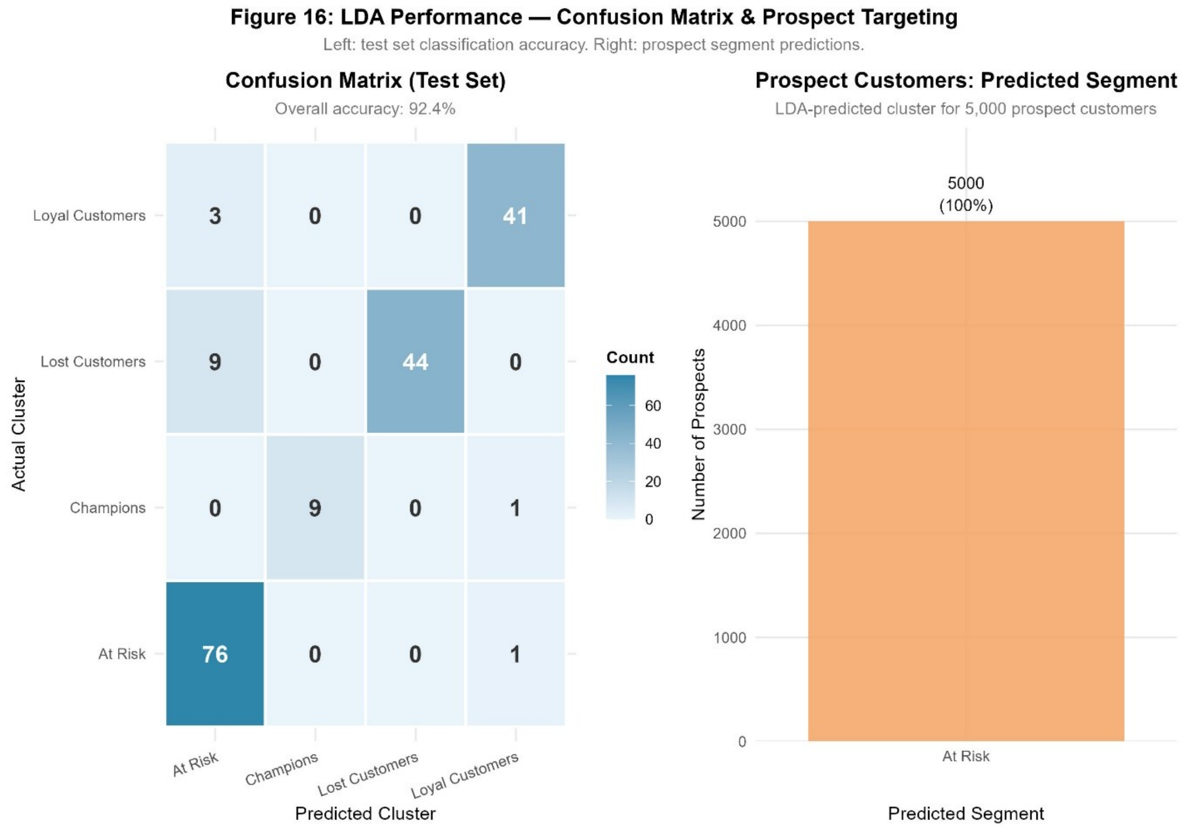
Target: Loyal Customers

The new private-label coffee product was chosen as the main target for the loyalty customers based on five converging lines of evidence. First, they are a commercially viable size, at 220 customers they are large enough to support a product launch, but small enough to allow for a very specific marketing strategy. Secondly, their overall contribution to total revenue is £32,392, which shows their willingness to spend. Third, they are very receptive to new product introductions because they are still in an active relationship with the retailer, and they are active and frequent. Fourth, they are in a strategic growth position: they are not yet Champions, but the behavioural gap is bridgeable and a well-designed product that fits their spending profile could move a significant percentage of this segment into the Champion level of engagement. Fifth, Champions were intentionally not included as marketing new products to already highly engaged customers can lead to cannibalisation of existing spend instead of incremental growth (Kotler and Keller, 2016).

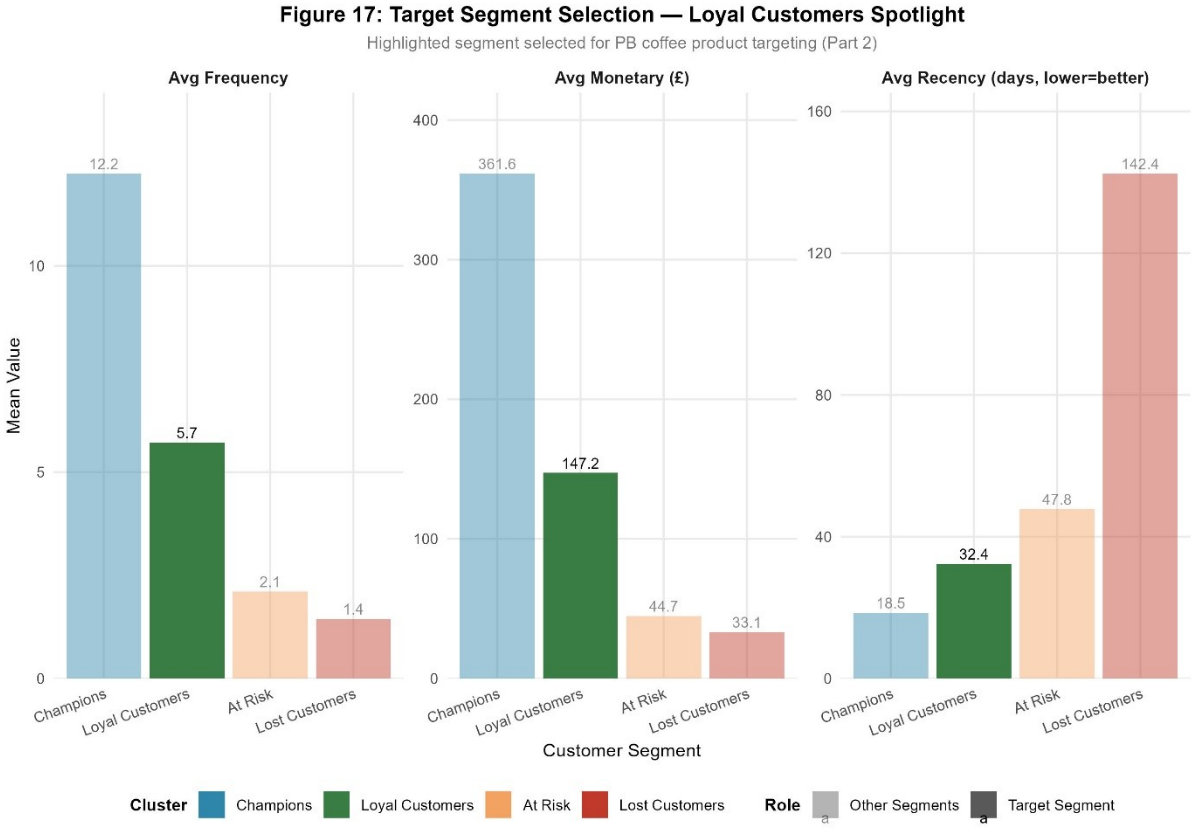


The cluster solution was validated using Linear Discriminant Analysis (LDA) with nine predictors, with a training accuracy of 94.31% and a test set accuracy of 92.39% (Kappa = 0.887). To ensure that the four

segments were not an artefact of the clustering algorithm, one-way ANOVA was used to confirm that all three RFM dimensions differed significantly across clusters (Frequency: F=503.2, p<0.001; Monetary: F=380.6, p<0.001; Recency: F=25.4, p<0.001).



Actual \ Predicted	At Risk	Champions	Lost Customers	Loyal Customers
Loyal Customers	3	0	0	41
Lost Customers	9	0	44	0
Champions	0	9	0	1
At Risk	76	0	0	1



4. Conjoint Analysis

4.1 Methodology

Once the target segment Loyal Customers was identified, the next analytical challenge was to find out what type of coffee product would be most attractive to them. This meant that the data from transactions was not enough and it was necessary to ask consumers directly about their preferences, which is the most suitable method for this type of analysis: conjoint analysis. Conjoint analysis is a survey-based method that shows respondents a series of product profiles, each with a different set of attribute levels, and asks them to rate or rank each profile. The method examines the variation of ratings with the variation of attributes, and breaks down the overall preference for the product into the contribution of each attribute, which is called part-worth utilities (Green and Srinivasan, 1990). The important point to remember is that consumers don't consider product features individually, they make trade-offs. A customer may be willing to pay more for a single-origin coffee that is ethically grown, but not for an instant coffee. These trade-offs are exactly what conjoint analysis is able to capture that simpler surveys are not.

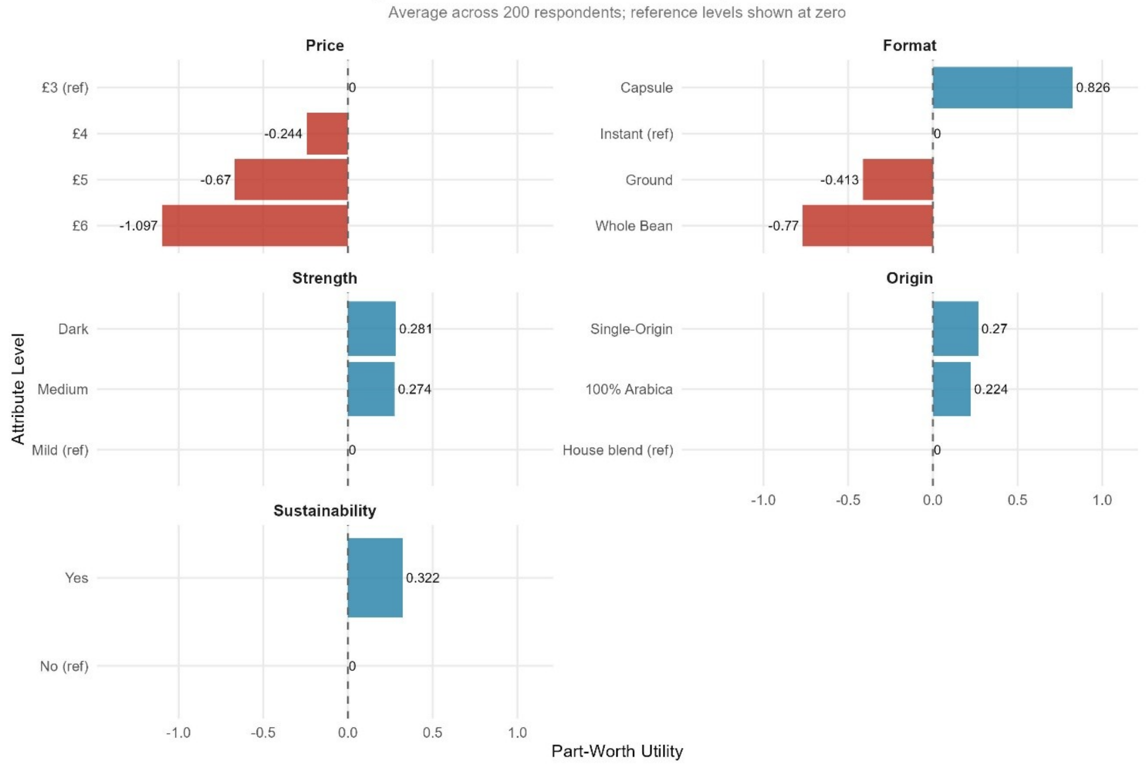
The five product attributes surveyed were price (£3, £4, £5, or £6), format (instant, capsule, ground, or whole bean), roast strength (mild, medium, or dark), origin (house blend, 100% Arabica blend, or single-

origin), and sustainability claim (yes or no). If all the attribute levels were combined in a full factorial design, there would be 4×4×3×3×2 = 288 product profiles, which would be too many for any one respondent to meaningfully evaluate. A fractional factorial design was therefore used, with 16 orthogonal profiles that allowed for the estimation of all part-worths independently, but were cognitively manageable for respondents. There are 12 parameters (5 attributes and 1 intercept) and 16 profiles, leaving 4 residual degrees of freedom, the minimum required for a statistically adequate design. The 200 respondents each gave a rating on a seven-point purchase likelihood scale for all 16 profiles, for a total of 3,200 observations.

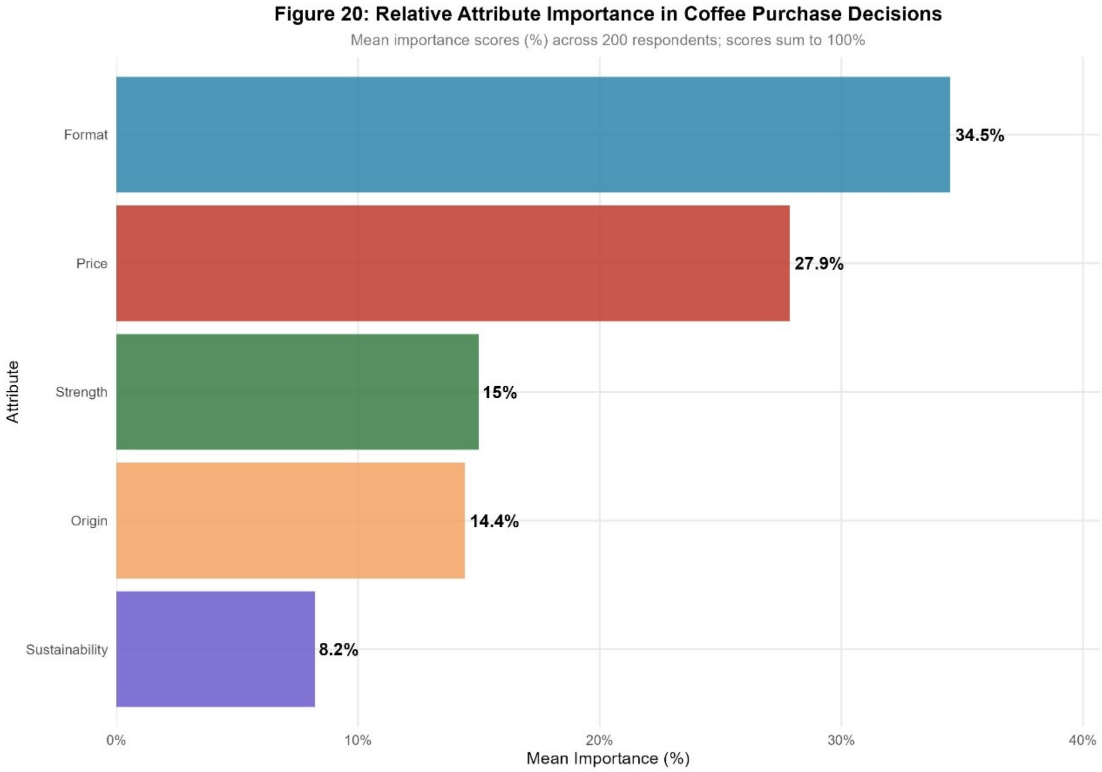
The individual-level part-worths were obtained by fitting an ordinary least squares regression to the 16 ratings of each of the 200 respondents, using dummy-coded levels of the attributes. The reference category was the lowest level of each attribute: price (£3), format (Instant), strength (Mild), origin (House blend), and sustainability (No). The mean part-worths for all respondents were then used to calculate the aggregate willingness to pay (WTP). This aggregate approach was chosen on purpose because some respondents may have very small price slopes, resulting in very high WTP values at the individual level which can affect the market-level average (Orme, 2006). Importance scores were calculated by normalising the range of part-worths for each attribute, with the total of all five attributes equal to 100%.

4.2 Conjoint Analysis Results

Figure 18: Mean Part-Worth Utilities by Attribute Level



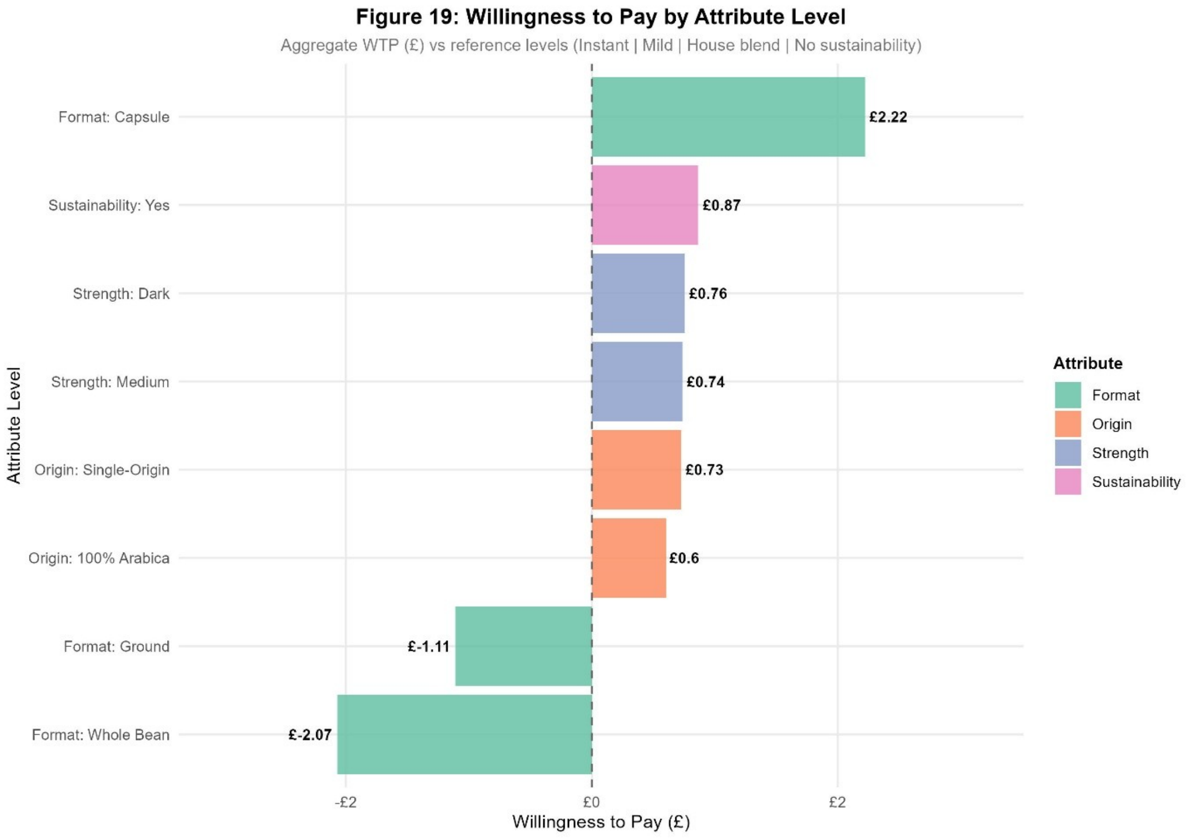
The results showed a distinct order of consumer preferences. The most significant attribute was format (34.5% of total purchase decision weight), followed by price (27.9%), strength (15.0%), origin (14.4%) and sustainability (8.2%).



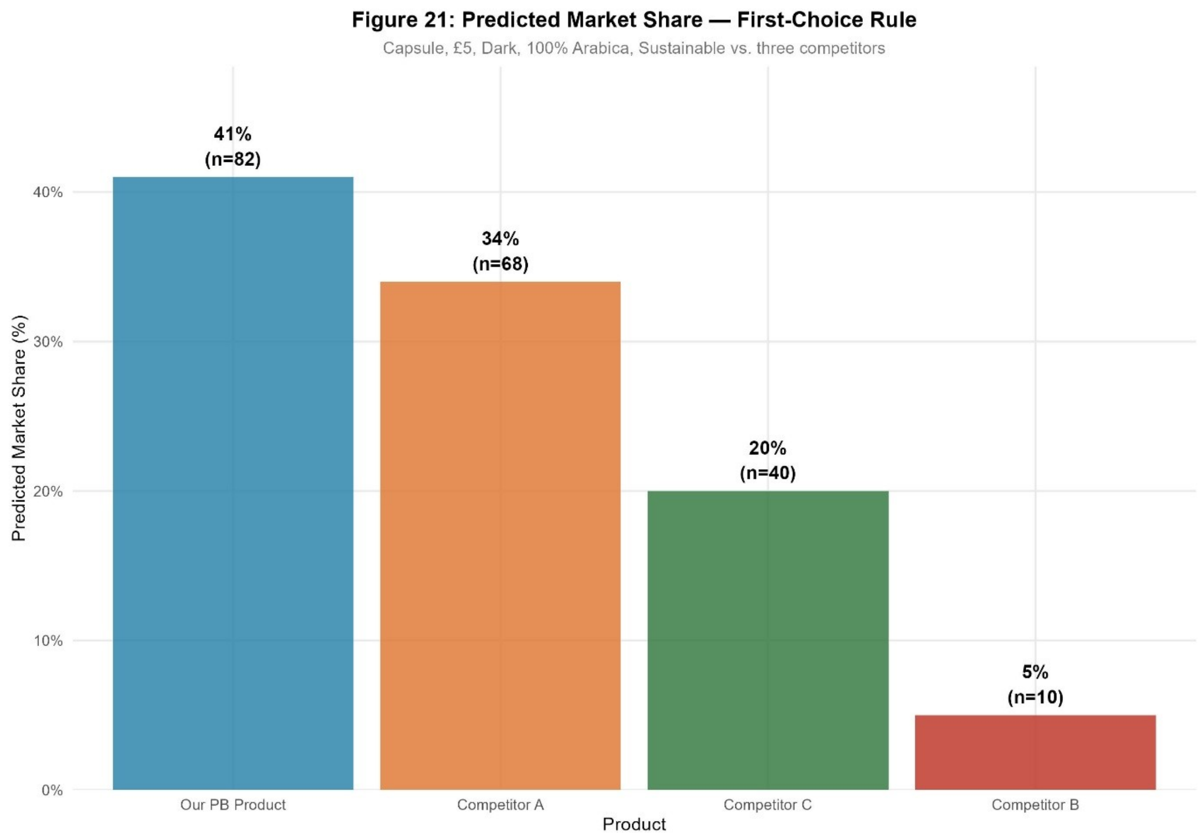
This is a commercially critical finding: the physical format of the product is more important to consumers than any other single decision, even price.

Capsule format had the highest part-worth utility (+0.826) compared to the instant reference, which equated to a willingness to pay premium of £2.22, the highest WTP of any attribute level in the analysis. Both whole bean and ground formats were penalised compared to instant, with WTP values of −£2.07 and −£1.11 respectively, indicating that these formats actively decrease purchase likelihood for this sample. As predicted, part-worths were monotonically negative on price: £4 (−0.244), £5 (−0.670), and £6 (−1.098) suggesting that consumers are always price sensitive, but the relatively low slope suggests that price is not the only determinant of choice. Sustainability was valued at +0.322 and WTP of £0.87, indicating that consumers value ethical credentials, but not at the cost of format or price. Dark roast (+£0.76) and single-origin (+£0.73) both enjoyed significant premiums

over their reference prices.



A market share simulation was performed based on the first-choice rule, which assumes that each respondent will select the product that is most likely to have the highest utility. The proposed private label product (capsule, £5, dark roast, 100% Arabica, sustainability claim) was predicted to have a market share of 41%, while Competitor A (£4, medium, house blend, no sustainability) was predicted to have a market share of 34%, Competitor C (£5, mild, single-origin, sustainability) was predicted to have a market share of 20%, and Competitor B (£6, dark, Arabica, no sustainability) was predicted to have a market share of only 5%. The proposed product leads, even though it is priced at £5, which is higher than the lowest competitor, is a good indicator that the attribute combination is providing enough utility to the consumer to warrant the price point.



5. Positioning Analysis

5.1 Methodology

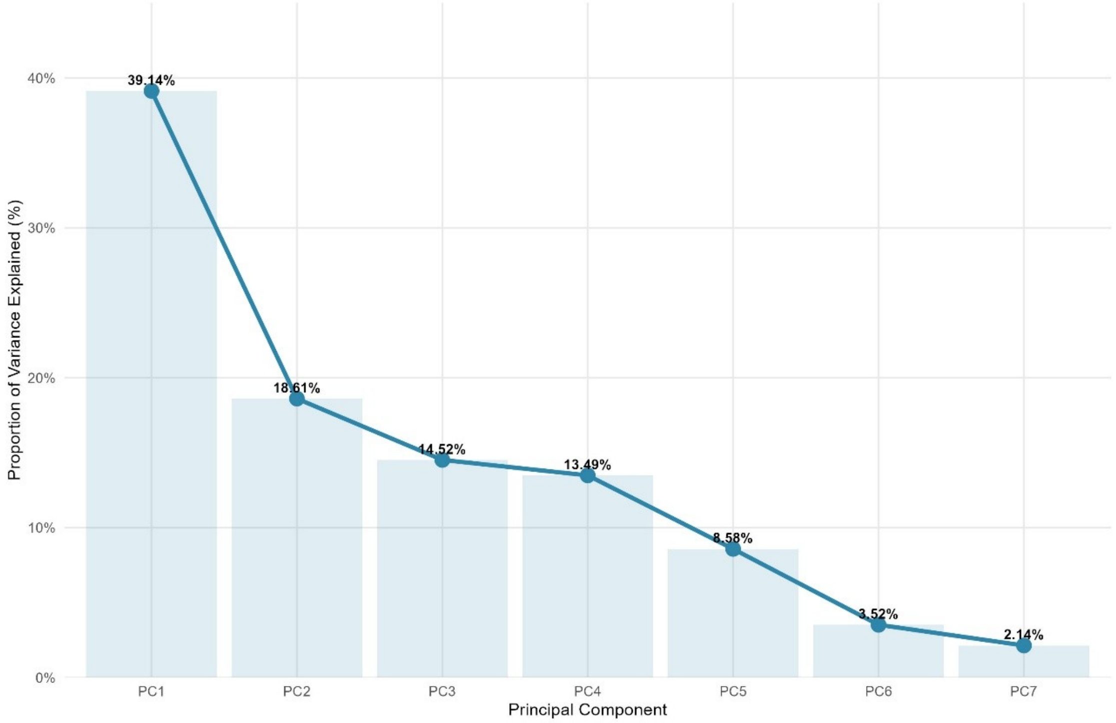
The last analytical step was to understand the positioning of the proposed private-label product in consumers' minds, in relation to the existing competitors. Positioning analysis is designed to show where products are placed in the minds of consumers, not by their actual attributes, but by the space they occupy in the mind. It identifies the filled, the empty, and the space where a new product can most effectively create a unique place in the mind of the consumer (Ries and Trout, 2001).

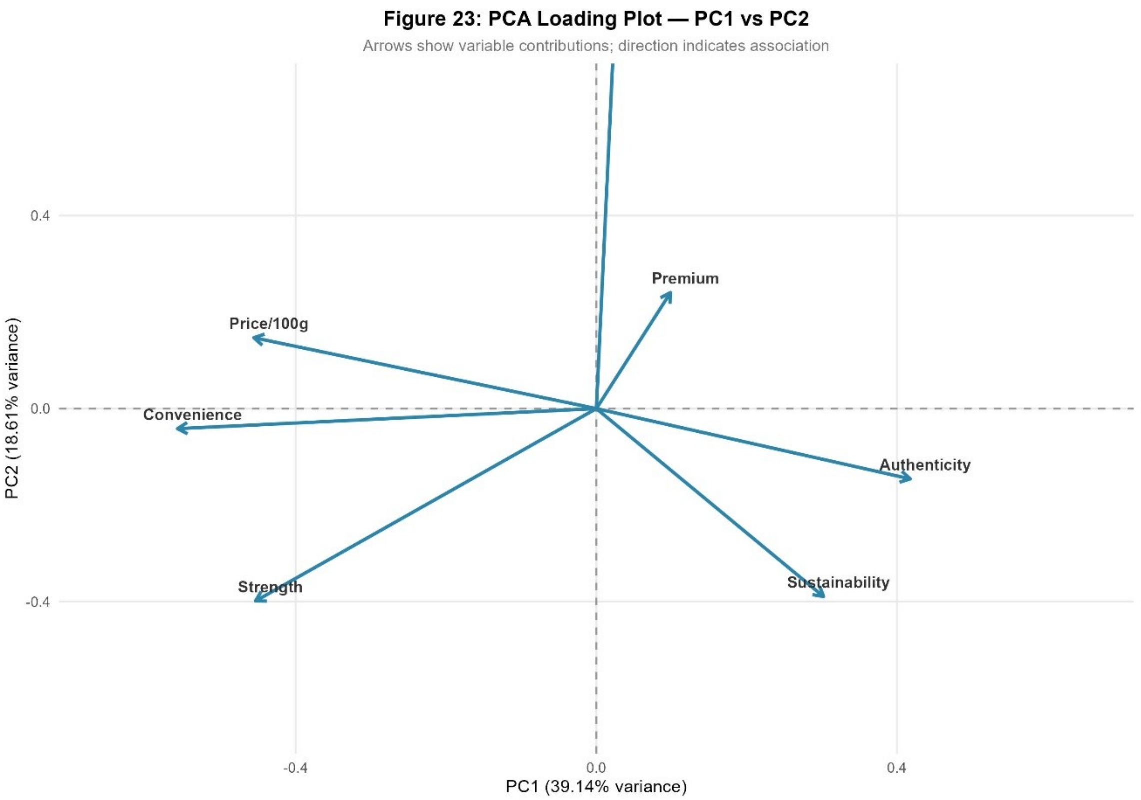
The perceptual map was created using Principal Component Analysis (PCA). PCA is a dimensionality reduction method that reduces the number of correlated variables to a smaller number of uncorrelated variables, called principal components, each of which represents a different dimension of variation in the data (Jolliffe, 2002). The input data set consisted of 24 already available coffee products, with seven perceptual and objective variables: price per 100g, strength level, convenience, authenticity, premium perception, perceived sustainability, and taste quality. All seven variables were standardized to have a mean of zero and a standard deviation of one, so that no single variable would dominate the components just because of the scale.

The first two principal components accounted for 57.8% of the total variance (PC1 39.1% and PC2 18.6%). PC1 was interpreted as a Mainstream versus Artisan axis: products loading negatively — high convenience, high strength, high price per 100g — are towards the mainstream, mass-market pole, and products loading positively on authenticity and sustainability are towards the artisan, craft pole. PC2 was considered as a Taste and Sensory Quality axis, which was mainly related to the Taste Quality scores. Together these two dimensions offer a meaningful and interpretable understanding of the competitive coffee market.

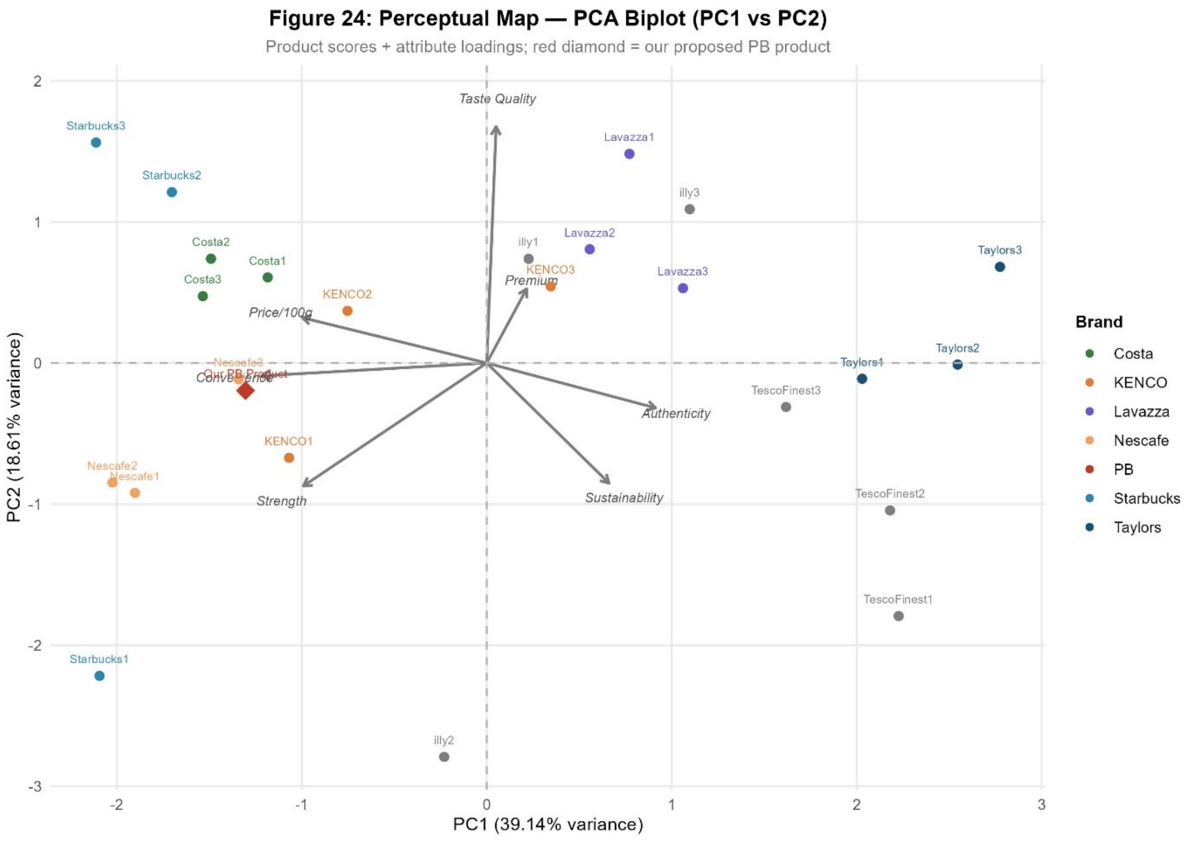
Figure 22: PCA Scree Plot — Proportion of Variance Explained

Each bar/point shows the % variance captured by each principal component





5.2 Positioning Discussion



The 24 products in the market were plotted on the perceptual map (PC1-PC2) and a clear competitive structure was formed. The mass-market

brands (Nescafé) grouped in the high-convenience, low-authenticity quadrant (mainstream pole of PC1), while the artisan and single-origin brands (Taylors and some Lavazza variants) were grouped towards the high-authenticity, high-sustainability pole. The taste quality axis (PC2) also distinguished products that were considered to be high quality sensory products from those that were considered to be convenient and accessible.

The proposed private label product (dark roast, £5, 100% Arabica, sustainability claim) was plotted on the perceptual map with the assumed attribute scores: price per 100g of £6.50, strength level 7, convenience 6.5, authenticity 4.0, premium 5.0, perceived sustainability 5.5, and taste quality 5.5. This positioned the PB product in a very strategic place: moderately high on the mainstream-convenience axis, due to the ease of use of the capsule format, and also meaningfully differentiated by the sustainability aspect and the Arabica origin, moving away from the purely commodity end of the market.

This positioning is commercially compelling for three reasons. First, it does not compete head-to-head with the most established mass-market competitors, who have the cost and distribution advantages of scale that would be hard to beat. Second, it is believable for a private-label product; consumers are more willing to try retailer own-brands in the premium-sustainability segment, if the product quality signals support the positioning (Ailawadi and Harlam, 2004). Third, it matches exactly the profile of the Loyal Customer target segment – a middle-aged, financially stable group who have proven to be willing to spend more than the category average, and a household profile that values convenience and quality credentials. The capsule format in particular caters to the convenience aspect that this segment is likely to value, and the sustainability and Arabica origin give the quality reassurance that makes this £5 price tag worth paying for compared to cheaper private label options.

6. Conclusion

The aim of this report was to provide a single, commercially relevant answer to the question: Who should a UK grocery retailer target with a new private label coffee product and what should it be like? Instead of using intuition or averages at the category level, the analysis was conducted in a sequential manner, with each quantitative method directly following the previous one. The analysis was conducted in a sequential manner, with each quantitative method building directly on the results of the previous one, rather than relying on intuition or category-level averages. The outcome is not four separate analyses, but a single, coherent argument that starts with the understanding of the

customers, their needs, and then the defensible position of a new product in the market.

The EDA stage set the groundwork for all the other stages. Once the data was cleaned to 4,962 transactions from 922 unique customers, the data showed that the shoppers were a mature age group (mean age 55 years), financially stable (mean income £68,900) and widely spread across educational levels. This demographic profile was not only a comfort check on the quality of the data, but it was a prerequisite for generalising the behavioural segmentation model to the individuals who have never transacted with the retailer.

Based on the RFM metrics, four customer behaviour segments were identified through cluster analysis. The strategically optimal target proved to be the Loyal Customers segment (n=220) who buy regularly (5.72 orders per person on average), spend meaningfully (£147 per person on average) and are actively engaged (32.4 days per person on average). Most importantly, they are not yet Champions (there is clear scope for more engagement) and their total revenue contribution of £32,392 demonstrates that they are commercially significant enough to warrant a targeted product launch. The LDA model was then validated on a held-out test set with 92.39% accuracy (Kappa = 0.887), and extended to all 5,000 prospect customers, providing the retailer with a list of prospects ready for action.

Conjoint analysis changed the question from who to what. It broke down the purchase likelihood into the independent contribution of each attribute for 200 respondents and found that format was the most important factor in choice (34.5% importance), with capsule commanding a willingness-to-pay premium of £2.22 over the instant reference. The market share simulation showed that the proposed product (capsule, £5, dark roast, 100% Arabica, sustainability claim) would take an estimated 41% of the simulated competitive market, outperforming all three modelled competitors despite not being the lowest cost product. As Green and Srinivasan (1990) note, conjoint analysis is particularly well suited to measuring the trade-offs consumers make between attributes, and is therefore the most suitable method available for making new product design decisions of this type.

PCA then placed the proposed product in the current competitive environment. The seven perceptual variables were reduced to two interpretable dimensions: a Mainstream vs Artisan axis (PC1, 39.1%) and a Taste Quality axis (PC2, 18.6%) and the proposed PB product was found to be in a strategically coherent space, being accessible enough in terms of convenience to not be niche, but also meaningfully differentiated by its sustainability credentials and Arabica origin. The usefulness of segmentation is that it uncovers the heterogeneity of consumer needs and helps the firm to move into a position where it can offer something

that is credibly different from the incumbents (Wedel and Kamakura 2000).

There are some limitations to the analysis. The part-worth estimates are not necessarily representative of the preferences of the intended target group, as the conjoint survey was not conducted with the Loyal Customer segment specifically. Furthermore, the perceptual attribute scores which were used to position the PB product on the perceptual map were assumed and not measured, which added a subjective element to the positioning analysis. In future, both of these conclusions would be reinforced by a segment-specific conjoint study and a dedicated brand perception survey.

In conclusion, this report shows that with careful, data-informed marketing analytics, there is no need to rely on guesswork when making a product launch decision. The target segment is well defined, the product is positioned based on empirically proven consumer trade-offs, and the positioning does not directly compete with scale players, but rather builds on the quality and sustainability signals that this segment is willing to pay for. These results give the retailer a solid, analytical basis for a private label coffee program that is truly customer-centric and not just based on category norms.

Declaration of usage of A.I.

In completing this assignment, Claude (Anthropic) was used as an assistive tool in accordance with Queen's University Belfast's Amber AI policy, which permits the use of generative AI in a supporting capacity where the substantive academic work remains the student's own. All R code, analytical decisions, interpretation of results, and written content were independently produced by me. Claude was consulted solely for assistance with structuring and drafting written sections of the report.

Reference List

Ailawadi, K.L. and Harlam, B. (2004) 'An empirical analysis of the determinants of retail margins: the role of store-brand share', *Journal of Marketing*, 68(1), pp. 147–165.

Everitt, B.S., Landau, S., Leese, M. and Stahl, D. (2011) *Cluster Analysis*. 5th edn. Chichester: Wiley.

Green, P.E. and Srinivasan, V. (1990) 'Conjoint analysis in marketing: new developments with implications for research and practice', *Journal of Marketing*, 54(4), pp. 3–19.

Hair, J.F., Black, W.C., Babin, B.J. and Anderson, R.E. (2019) *Multivariate Data Analysis*. 8th edn. Cengage Learning.

Hughes, A.M. (1994) *Strategic Database Marketing*. Chicago: Probus Publishing.

Jolliffe, I.T. (2002) *Principal Component Analysis*. 2nd edn. New York: Springer.

Kotler, P. and Keller, K.L. (2016) *Marketing Management*. 15th edn. Harlow: Pearson Education.

Orme, B.K. (2006) *Getting Started with Conjoint Analysis: Strategies for Product Design and Pricing Research*. Madison: Research Publishers.

Ries, A. and Trout, J. (2001) *Positioning: The Battle for Your Mind*. New York: McGraw-Hill.

Tukey, J.W. (1977) *Exploratory Data Analysis*. Reading, MA: Addison-Wesley.

Wedel, M. and Kamakura, W.A. (2000) *Market Segmentation: Conceptual and Methodological Foundations*. 2nd edn. Boston: Kluwer Academic Publishers.

Appendix

```
# ==========================================================================================
# ITAO7106 Marketing Analytics — Queen's University Belfast
# Student Name   : [Jai Tajvir Baloda]
# Student ID     : [40485958]
# Assignment     : Market STP Strategy and 4P Marketing Mix
# Submission Date: 15th May 2026
# ==========================================================================================
```

```
#
==============================================================
===========================

# SECTION 0: PACKAGE INSTALLATION AND LIBRARY LOADING

#
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===========================


# I installed and loaded all libraries required across the entire assignment

# in a single unified block at the outset. This ensures all dependencies are

# available before any analysis begins and avoids redundant install/library

# calls throughout the script.


if (!require("tidyverse"))    install.packages("tidyverse")

if (!require("lubridate"))    install.packages("lubridate")

if (!require("scales"))       install.packages("scales")

if (!require("ggcorrplot"))   install.packages("ggcorrplot")

if (!require("RColorBrewer")) install.packages("RColorBrewer")

if (!require("patchwork"))    install.packages("patchwork")

if (!require("knitr"))        install.packages("knitr")

if (!require("gridExtra"))    install.packages("gridExtra")

if (!require("cluster"))      install.packages("cluster")

if (!require("factoextra"))   install.packages("factoextra")

if (!require("dendextend"))   install.packages("dendextend")

if (!require("MASS"))         install.packages("MASS")

if (!require("caret"))        install.packages("caret")

if (!require("fmsb"))         install.packages("fmsb")


library(tidyverse)

library(lubridate)

library(scales)

library(ggcorrplot)

library(RColorBrewer)

library(patchwork)
```

```
library(knitr)
library(gridExtra)
library(cluster)
library(factoextra)
library(dendextend)
library(MASS)
library(caret)
library(fmsb)

# I overrode the select() conflict caused by MASS masking dplyr::select,
# ensuring all subsequent select() calls use the dplyr version throughout.
select <- dplyr::select

# ===============================================================================
# PART 1 — STEP 1: EXPLORATORY DATA ANALYSIS (EDA)
# Datasets: All five datasets loaded, cleaned, and inspected here.
#           Retail + Prospect are the primary datasets for Part 1 analysis.
#           Conjoint, Product Profiles, and Product Attributes are cleaned
#           here and saved for use in Part 2.
# ===============================================================================

# ---------------------------------------------------------------------------
# SECTION 1: DATA LOADING — ALL FIVE DATASETS
# ---------------------------------------------------------------------------

# I loaded all five datasets at the outset so that each could be inspected,
# cleaned, and saved in a single, self-contained EDA script.

# Part 1 datasets
```

```r
retail_raw    <- read_csv("C:/Users/Jai Baloda/Desktop/Jai DELL/Masters Business Analytics/Semester 2/Marketing Analytics/Retail transaction data.csv", show_col_types = FALSE)

prospect_raw      <- read_csv("C:/Users/Jai Baloda/Desktop/Jai DELL/Masters Business Analytics/Semester 2/Marketing Analytics/Prospect customers info.csv",       show_col_types = FALSE)


# Part 2 datasets

conjoint_raw      <- read_csv("C:/Users/Jai Baloda/Desktop/Jai DELL/Masters Business Analytics/Semester 2/Marketing Analytics/Conjoint survey results-1.csv",     show_col_types = FALSE)

profiles_raw      <- read_csv("C:/Users/Jai Baloda/Desktop/Jai DELL/Masters Business Analytics/Semester 2/Marketing Analytics/Product profiles.csv", show_col_types = FALSE)

prod_attr_raw    <- read_csv("C:/Users/Jai Baloda/Desktop/Jai DELL/Masters Business Analytics/Semester 2/Marketing Analytics/Product attributes information.csv",show_col_types = FALSE)


# I confirmed the dimensions of each dataset to verify successful loading.

cat("=== Dataset Dimensions ===\n")

cat("Retail transaction data      :", dim(retail_raw),    "\n")

cat("Prospect customers info      :", dim(prospect_raw),  "\n")

cat("Conjoint survey results      :", dim(conjoint_raw),  "\n")

cat("Product profiles             :", dim(profiles_raw),  "\n")

cat("Product attributes info      :", dim(prod_attr_raw), "\n")


# --------------------------------------------------------------------------

# SECTION 2: INITIAL INSPECTION — STRUCTURE AND MISSING VALUES

# --------------------------------------------------------------------------


cat("\n=== RETAIL: Structure ===\n");    glimpse(retail_raw)

cat("\n=== PROSPECT: Structure ===\n");  glimpse(prospect_raw)

cat("\n=== CONJOINT: Structure ===\n");  glimpse(conjoint_raw)

cat("\n=== PROFILES: Structure ===\n");  glimpse(profiles_raw)

cat("\n=== PROD ATTR: Structure ===\n"); glimpse(prod_attr_raw)
```

```
cat("\n=== Missing Values Summary ===\n")
cat("Retail:\n");    print(colSums(is.na(retail_raw)))
cat("Prospect:\n");  print(colSums(is.na(prospect_raw)))
cat("Conjoint:\n");  print(colSums(is.na(conjoint_raw)))
cat("Profiles:\n");  print(colSums(is.na(profiles_raw)))
cat("Prod Attr:\n"); print(colSums(is.na(prod_attr_raw)))

cat("\n=== Duplicate Rows ===\n")
cat("Retail duplicates    :", sum(duplicated(retail_raw)),    "\n")
cat("Prospect duplicates  :", sum(duplicated(prospect_raw)),  "\n")
cat("Conjoint duplicates  :", sum(duplicated(conjoint_raw)),  "\n")
cat("Profiles duplicates  :", sum(duplicated(profiles_raw)),  "\n")
cat("Prod Attr duplicates :", sum(duplicated(prod_attr_raw)), "\n")

# ---------------------------------------------------------------------------
# SECTION 3: DATA CLEANING
# ---------------------------------------------------------------------------

# I defined a reusable outlier-capping (Winsorisation) function using a 3xIQR
# fence. Rather than deleting outlier rows, I capped extreme values to preserve
# all valid transaction records while reducing the distorting effect of extremes.
cap_outliers <- function(x, multiplier = 3) {
  Q1      <- quantile(x, 0.25, na.rm = TRUE)
  Q3      <- quantile(x, 0.75, na.rm = TRUE)
  IQR_val <- Q3 - Q1
  lower   <- Q1 - multiplier * IQR_val
  upper   <- Q3 + multiplier * IQR_val
  pmin(pmax(x, lower), upper)
}

# I also defined a helper function to apply consistent demographic factor
# labelling across both the retail and prospect datasets.
label_demographics <- function(df) {
```

```
  df %>%
    mutate(
      Married = factor(Married,
                       levels = c(0, 1),
                       labels = c("Not Married", "Married")),
      Education = factor(Education,
                         levels = c(1, 2, 3),
                         labels = c("No University Degree",
                                    "Undergraduate",
                                    "Postgraduate")),
      Work = factor(Work,
                    levels = 1:11,
                    labels = c("Health Services",
                               "Financial Services",
                               "Sales",
                               "Advertising / PR",
                               "Education",
                               "Construction / Logistics",
                               "Engineering",
                               "Technology",
                               "Retailing / Services",
                               "SME / Self-Employed",
                               "Transportation"))
    )
}

# ---- 3a. RETAIL TRANSACTION DATA CLEANING ----

# Step 1: I removed the two fully duplicated rows, as these were exact repeat
# entries of the same transaction and would artificially inflate spending totals.
retail_clean <- retail_raw %>% distinct()
cat("\nAfter removing duplicates       :", nrow(retail_clean), "rows\n")
```

```
# Step 2: I removed the 36 rows where ProductID was recorded as 'PNA'
# (not available). These rows had no valid product identifier or category
# and would contribute nothing to category-level or RFM analyses.
retail_clean <- retail_clean %>% filter(ProductID != "PNA")
cat("After removing PNA ProductIDs   :", nrow(retail_clean), "rows\n")

# Step 3: I removed the remaining rows with missing ProductCategory values.
retail_clean <- retail_clean %>% filter(!is.na(ProductCategory))
cat("After removing missing Category :", nrow(retail_clean), "rows\n")

# Step 4: I converted InvoiceDate to a proper Date object.
retail_clean <- retail_clean %>%
  mutate(InvoiceDate = as.Date(InvoiceDate, format = "%Y-%m-%d"))

# Step 5: I created a TotalSpend variable as UnitPrice x Quantity.
retail_clean <- retail_clean %>%
  mutate(TotalSpend = UnitPrice * Quantity)

# Step 6: I applied Winsorisation (3xIQR capping) to UnitPrice and Quantity.
# UnitPrice had a maximum of £1,592 against a mean of £4.13, and Quantity
# reached 1,930 against a mean of approximately 10 — both far beyond
# plausible single-item grocery transaction values.
retail_clean <- retail_clean %>%
  mutate(
    UnitPrice  = cap_outliers(UnitPrice),
    Quantity   = cap_outliers(Quantity),
    TotalSpend = UnitPrice * Quantity      # Recalculated after capping
  )

# Step 7: I converted all categorical code columns to labelled factor variables.
retail_clean <- label_demographics(retail_clean)
cat("Final retail_clean dimensions   :", dim(retail_clean), "\n")
```

```
# ---- 3b. PROSPECT CUSTOMERS DATA CLEANING ----

# I inspected the prospect dataset and found no missing values, no duplicates,
# and no values outside the valid ranges for any variable. No rows were removed.
prospect_clean <- prospect_raw %>% label_demographics()
cat("Final prospect_clean dimensions :", dim(prospect_clean), "\n")

# ---- 3c. CONJOINT SURVEY DATA CLEANING ----

# I verified that the conjoint dataset contained no missing values, no
# duplicates, and that all 200 respondents rated all 16 product profiles.
conjoint_clean <- conjoint_raw %>%
  mutate(
    format         = factor(format,
                            levels = c("Instant", "Capsule", "Ground", "Whole bean")),
    strength       = factor(strength,
                            levels = c("Mild", "Medium", "Dark"),
                            ordered = TRUE),
    origin         = factor(origin,
                            levels = c("House blend",
                                       "100% Arabica blend",
                                       "Single-origin")),
    sustainability = factor(sustainability, levels = c("No", "Yes"))
  )
cat("Final conjoint_clean dimensions :", dim(conjoint_clean), "\n")

# ---- 3d. PRODUCT PROFILES DATA CLEANING ----

# I confirmed that the product profiles dataset was fully clean — no missing
# values, no duplicates, and exactly 16 orthogonal profiles.
profiles_clean <- profiles_raw %>%
  mutate(
```

```
    format        = factor(format,
                           levels = c("Instant", "Capsule", "Ground", "Whole bean")),
    strength      = factor(strength,
                           levels = c("Mild", "Medium", "Dark"),
                           ordered = TRUE),
    origin        = factor(origin,
                           levels = c("House blend",
                                      "100% Arabica blend",
                                      "Single-origin")),
    sustainability = factor(sustainability, levels = c("No", "Yes"))
  )
cat("Final profiles_clean dimensions :", dim(profiles_clean), "\n")

# ---- 3e. PRODUCT ATTRIBUTES DATA CLEANING ----

# I corrected the column name typo 'sustaintability_claim' (extra 'n') to
# 'sustainability_claim'. The low taste_quality score for Starbucks1 (0.239)
# was retained as a genuine low-rated data point after outlier inspection.
prod_attr_clean <- prod_attr_raw %>%
  rename(sustainability_claim = sustaintability_claim) %>%
  mutate(
    brand               = str_extract(product_name, "^[A-Za-z]+"),
    is_decaf            = factor(is_decaf,
                                 levels = c(0, 1),
                                 labels = c("Not Decaf", "Decaf")),
    sustainability_claim = factor(sustainability_claim,
                                 levels = c(0, 1),
                                 labels = c("No Claim", "Sustainability Claim"))
  )
cat("Final prod_attr_clean dimensions:", dim(prod_attr_clean), "\n")

# ---------------------------------------------------------------------------
# SECTION 4: DESCRIPTIVE STATISTICS
```

```
# ---------------------------------------------------------------------------

cat("\n=== RETAIL: Descriptive Statistics ===\n")
retail_clean %>%
  select(UnitPrice, Quantity, TotalSpend, Income, Age, HouseholdSize) %>%
  summary() %>% print()

cat("\n=== PROSPECT: Descriptive Statistics ===\n")
prospect_clean %>%
  select(Income, Age, HouseholdSize) %>%
  summary() %>% print()

cat("\n=== CONJOINT: Rating and Price Statistics ===\n")
conjoint_clean %>%
  select(price, rating) %>%
  summary() %>% print()

cat("\n=== PRODUCT ATTRIBUTES: Perceptual Rating Statistics ===\n")
prod_attr_clean %>%
  select(price, price_100g, strength_level,
         convenience, authenticity, premium,
         perceived_sustainability, taste_quality) %>%
  summary() %>% print()

cat("\n--- Key Counts ---\n")
cat("Retail: Unique customers      :", n_distinct(retail_clean$CustomerID),   "\n")
cat("Retail: Unique invoices       :", n_distinct(retail_clean$InvoiceNo),     "\n")
cat("Retail: Date range            :",
    as.character(min(retail_clean$InvoiceDate)), "to",
    as.character(max(retail_clean$InvoiceDate)), "\n")
cat("Retail: Product categories    :", n_distinct(retail_clean$ProductCategory), "\n")
```

```
cat("Conjoint: Respondents        :", n_distinct(conjoint_clean$respondent_id),  "\n")

cat("Conjoint: Profiles per person  :", n_distinct(conjoint_clean$productNo),     "\n")

cat("Product attributes: Brands    :", n_distinct(prod_attr_clean$brand),       "\n")


# ---------------------------------------------------------------------------

# SECTION 5: VISUALISATIONS — EDA (Figures 1–8)

# ---------------------------------------------------------------------------


# I defined a consistent custom ggplot theme to ensure a professional and

# uniform visual style is applied to all plots throughout this assignment.

theme_assignment <- function() {

  theme_minimal(base_size = 12) +

    theme(

      plot.title       = element_text(face = "bold", size = 13, hjust = 0.5),

      plot.subtitle    = element_text(size = 10, hjust = 0.5, colour = "grey45"),

      axis.title       = element_text(size = 11),

      axis.text        = element_text(size = 9),

      legend.title     = element_text(size = 10, face = "bold"),

      legend.text      = element_text(size = 9),

      panel.grid.minor = element_blank(),

      strip.text       = element_text(face = "bold", size = 10)

    )

}


# ---- FIGURE 1: Monthly Transaction Volume Over Time ----

# I plotted the number of transactions per month to understand temporal

# purchasing patterns across the January to August 2025 observation window.


monthly_vol <- retail_clean %>%

  mutate(Month = floor_date(InvoiceDate, "month")) %>%

  count(Month, name = "Transactions")
```

```
p1 <- ggplot(monthly_vol, aes(x = Month, y = Transactions)) +
  geom_area(fill = "#2E86AB", alpha = 0.15) +
  geom_line(colour = "#2E86AB", linewidth = 1.3) +
  geom_point(colour = "#2E86AB", size = 3) +
  scale_x_date(date_labels = "%b %Y", date_breaks = "1 month") +
  scale_y_continuous(labels = comma, limits = c(0, NA),
                     expand = expansion(mult = c(0, 0.1))) +
  labs(
    title    = "Figure 1: Monthly Transaction Volume (Jan–Aug 2025)",
    subtitle = "Number of transaction line items recorded per calendar month",
    x = "Month", y = "Number of Transactions"
  ) +
  theme_assignment() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
print(p1)

# ---- FIGURE 2: Revenue and Transaction Count by Product Category ----
# I created a dual-panel bar chart showing total revenue and transaction count
# by product category. This segment-wise view identifies the most valuable and
# most frequently purchased categories, supporting the business case for a
# new private-label coffee product.

cat_summary <- retail_clean %>%
  group_by(ProductCategory) %>%
  summarise(Revenue = sum(TotalSpend), Transactions = n(), .groups = "drop")

p2a <- ggplot(cat_summary,
              aes(x = reorder(ProductCategory, Revenue), y = Revenue)) +
  geom_col(fill = "#3A7D44", alpha = 0.85) +
  geom_text(aes(label = paste0("£", round(Revenue / 1000, 1), "k")),
            hjust = -0.1, size = 3) +
  coord_flip() +
```

```
  scale_y_continuous(labels = label_dollar(prefix = "£"),
                     expand = expansion(mult = c(0, 0.2))) +
  labs(title = "Total Revenue", x = NULL, y = "Revenue (£)") +
  theme_assignment()

p2b <- ggplot(cat_summary,
              aes(x = reorder(ProductCategory, Transactions), y = Transactions)) +
  geom_col(fill = "#E07B39", alpha = 0.85) +
  geom_text(aes(label = comma(Transactions)), hjust = -0.1, size = 3) +
  coord_flip() +
  scale_y_continuous(labels = comma, expand = expansion(mult = c(0, 0.2))) +
  labs(title = "Transaction Count", x = NULL, y = "No. of Transactions") +
  theme_assignment()

p2 <- p2a + p2b +
  plot_annotation(
    title    = "Figure 2: Revenue and Transaction Count by Product Category",
    subtitle = "Segment-wise view across all 10 product categories (Jan–Aug 2025)",
    theme    = theme(plot.title    = element_text(face = "bold", size = 13, hjust = 0.5),
                plot.subtitle = element_text(size = 10, hjust = 0.5, colour = "grey45")))
print(p2)

# ---- FIGURE 3: Customer Demographics — Age, Income, Household Size ----
# I produced a three-panel demographic overview for unique current customers
# to understand the distribution of key profiling variables.

cust_demo <- retail_clean %>% distinct(CustomerID, .keep_all = TRUE)

p3a <- ggplot(cust_demo, aes(x = Age)) +
  geom_histogram(aes(y = after_stat(density)), bins = 25,
                 fill = "#6A5ACD", colour = "white", alpha = 0.8) +
```

```
  geom_density(colour = "#3B2F8F", linewidth = 1.1) +
  labs(title = "Age Distribution", x = "Age (years)", y = "Density") +
  theme_assignment()

p3b <- ggplot(cust_demo, aes(x = Income)) +
  geom_histogram(aes(y = after_stat(density)), bins = 25,
                 fill = "#2E86AB", colour = "white", alpha = 0.8) +
  geom_density(colour = "#1A5276", linewidth = 1.1) +
  scale_x_continuous(labels = dollar_format(prefix = "£")) +
  labs(title = "Income Distribution", x = "Annual Income (£k)", y = "Density") +
  theme_assignment()

p3c <- cust_demo %>%
  count(HouseholdSize) %>%
  ggplot(aes(x = factor(HouseholdSize), y = n, fill = factor(HouseholdSize))) +
  geom_col(alpha = 0.85, show.legend = FALSE) +
  geom_text(aes(label = n), vjust = -0.4, size = 3.2) +
  scale_fill_brewer(palette = "Blues") +
  scale_y_continuous(expand = expansion(mult = c(0, 0.12))) +
  labs(title = "Household Size", x = "No. of People in Household", y = "Count") +
  theme_assignment()

p3 <- p3a + p3b + p3c +
  plot_annotation(
    title    = "Figure 3: Customer Demographics — Age, Income & Household Size",
    subtitle = "One record per unique current customer (n = 925)",
    theme    = theme(plot.title    = element_text(face = "bold", size = 13, hjust =
0.5),
                plot.subtitle = element_text(size = 10, hjust = 0.5, colour = "grey45")))
print(p3)

# ---- FIGURE 4: Education, Marital Status & Occupation ----
# I visualised three remaining demographic variables as a combined figure
```

```
# to support customer profiling and segment characterisation in Steps 3–4.

edu_dist <- cust_demo %>% count(Education) %>% mutate(Pct = n / sum(n) * 100)

p4a <- ggplot(edu_dist, aes(x = Education, y = Pct, fill = Education)) +
  geom_col(alpha = 0.85, show.legend = FALSE) +
  geom_text(aes(label = paste0(round(Pct, 1), "%")), vjust = -0.4, size = 3.2) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(expand = expansion(mult = c(0, 0.12))) +
  labs(title = "Education Level", x = NULL, y = "% of Customers") +
  theme_assignment() +
  theme(axis.text.x = element_text(angle = 18, hjust = 1))

married_dist <- cust_demo %>%
  count(Married) %>%
  mutate(Pct = n / sum(n) * 100, Label = paste0(Married, "\n", round(Pct, 1), "%"))

p4b <- ggplot(married_dist, aes(x = "", y = Pct, fill = Married)) +
  geom_col(width = 1, colour = "white", linewidth = 0.5) +
  coord_polar(theta = "y") +
  geom_text(aes(label = Label), position = position_stack(vjust = 0.5),
            size = 3.8, fontface = "bold", colour = "white") +
    scale_fill_manual(values = c("Not Married" = "#E07B39", "Married" = "#2E86AB")) +
  labs(title = "Marital Status", fill = NULL) +
  theme_void() +
  theme(plot.title  = element_text(face = "bold", size = 12, hjust = 0.5),
      legend.text = element_text(size = 9))

work_dist <- cust_demo %>%
  count(Work) %>% mutate(Pct = n / sum(n) * 100) %>% arrange(desc(Pct))
```

```
p4c <- ggplot(work_dist, aes(x = reorder(Work, Pct), y = Pct)) +
  geom_col(fill = "#3A7D44", alpha = 0.82) +
  geom_text(aes(label = paste0(round(Pct, 1), "%")), hjust = -0.1, size = 2.8) +
  coord_flip() +
  scale_y_continuous(expand = expansion(mult = c(0, 0.2))) +
  labs(title = "Occupation Sector", x = NULL, y = "% of Customers") +
  theme_assignment()

p4 <- (p4a | p4b) / p4c +
  plot_annotation(
    title    = "Figure 4: Education, Marital Status & Occupation of Current Customers",
    subtitle = "Segment-wise demographic breakdown (n = 925 unique customers)",
    theme    = theme(plot.title    = element_text(face = "bold", size = 13, hjust = 0.5),
                     plot.subtitle = element_text(size = 10, hjust = 0.5, colour = "grey45")))
print(p4)

# ---- FIGURE 5: Income — Current vs. Prospect Customers ----
# I overlaid the income density curves of current and prospective customers
# to assess demographic similarity. Close overlap supports LDA generalisation.

income_compare <- bind_rows(
  cust_demo      %>% select(Income) %>% mutate(Group = "Current Customers (n=925)"),
  prospect_clean %>% select(Income) %>% mutate(Group = "Prospect Customers (n=5,000)")
)

p5 <- ggplot(income_compare, aes(x = Income, fill = Group, colour = Group)) +
  geom_density(alpha = 0.30, linewidth = 1.1) +
  scale_fill_manual(values   = c("Current Customers (n=925)"    = "#2E86AB",
                                 "Prospect Customers (n=5,000)" = "#E07B39")) +
  scale_colour_manual(values = c("Current Customers (n=925)"    = "#2E86AB",
```

```
                          "Prospect Customers (n=5,000)" = "#E07B39")) +
  scale_x_continuous(labels = dollar_format(prefix = "£")) +
  labs(
    title    = "Figure 5: Income Distribution — Current vs. Prospect Customers",
      subtitle = "Overlapping density curves; close alignment supports LDA generalisation",
    x = "Annual Household Income (£k)", y = "Density", fill = NULL, colour = NULL
  ) +
  theme_assignment()
print(p5)

# ---- FIGURE 6: Segment-wise — Age by Product Category ----
# I created a boxplot showing customer age distributions across product
# categories, directly informing target segment selection for coffee.

p6 <- ggplot(retail_clean,
             aes(x = reorder(ProductCategory, Age, FUN = median),
                 y = Age, fill = ProductCategory)) +
  geom_boxplot(alpha = 0.75, outlier.size = 1, outlier.alpha = 0.4,
               show.legend = FALSE) +
  coord_flip() +
  scale_fill_brewer(palette = "Set3") +
  labs(
    title    = "Figure 6: Age Distribution by Product Category",
    subtitle = "Segment-wise boxplot — categories ordered by median customer age",
    x = "Product Category", y = "Customer Age (years)"
  ) +
  theme_assignment()
print(p6)

# ---- FIGURE 7: Segment-wise — Average Spend by Education & Marital Status ----
# I computed average spend broken down by education and marital status to
```

```
# identify high-spending demographic segments for strategic targeting.

avg_spend_seg <- retail_clean %>%
  group_by(Education, Married) %>%
  summarise(AvgSpend = mean(TotalSpend), .groups = "drop")

p7 <- ggplot(avg_spend_seg, aes(x = Education, y = AvgSpend, fill = Married)) +
  geom_col(position = "dodge", alpha = 0.85, colour = "white") +
  geom_text(aes(label = paste0("£", round(AvgSpend, 2))),
            position = position_dodge(width = 0.9), vjust = -0.4, size = 3.0) +
    scale_fill_manual(values = c("Not Married" = "#E07B39", "Married" =
"#2E86AB")) +
  scale_y_continuous(labels = label_dollar(prefix = "£"),
                     expand = expansion(mult = c(0, 0.14))) +
  labs(
    title    = "Figure 7: Average Spend per Transaction by Education & Marital
Status",
    subtitle = "Segment-wise view — interaction of education level and marital
status",
    x = "Education Level", y = "Average Spend per Transaction (£)", fill = "Marital
Status"
  ) +
  theme_assignment() +
  theme(axis.text.x = element_text(angle = 15, hjust = 1))
print(p7)

# ---- FIGURE 8: Part 2 Dataset Overview — Conjoint Ratings & Perceptual Scores
----
# I produced a two-panel overview of the Part 2 datasets. The left panel
# confirms conjoint ratings are well-distributed across the 1–7 scale.
# The right panel confirms adequate perceptual variance for PCA in Part 2.

p8a <- ggplot(conjoint_clean, aes(x = factor(rating))) +
  geom_bar(fill = "#6A5ACD", alpha = 0.85) +
```

```
  geom_text(stat = "count", aes(label = after_stat(count)), vjust = -0.4, size = 3.0) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.12))) +
  labs(title = "Conjoint Survey: Rating Distribution",
       x = "Purchase Likelihood Rating (1–7)", y = "Frequency") +
  theme_assignment()

prod_attr_long <- prod_attr_clean %>%
  select(product_name, convenience, authenticity,
         premium, perceived_sustainability, taste_quality) %>%
  pivot_longer(-product_name, names_to = "Attribute", values_to = "Score") %>%
  mutate(Attribute = str_to_title(str_replace_all(Attribute, "_", " ")))

p8b <- ggplot(prod_attr_long, aes(x = Attribute, y = Score, fill = Attribute)) +
  geom_boxplot(alpha = 0.75, outlier.size = 1.5, outlier.alpha = 0.5,
               show.legend = FALSE) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(limits = c(0, 7.5)) +
  labs(title = "Product Attributes: Perceptual Score Distributions",
       x = "Perceptual Attribute", y = "Customer Rating (1–7)") +
  theme_assignment() +
  theme(axis.text.x = element_text(angle = 20, hjust = 1))

p8 <- p8a + p8b +
  plot_annotation(
    title    = "Figure 8: Part 2 Dataset Overview — Conjoint Ratings & Product Perceptions",
    subtitle = paste0("Left: conjoint rating distribution (n=3,200 ratings). ",
                      "Right: perceptual attribute scores across 24 coffee products."),
    theme    = theme(plot.title    = element_text(face = "bold", size = 13, hjust = 0.5),
                plot.subtitle = element_text(size = 10, hjust = 0.5, colour = "grey45")))
print(p8)
```

```
# --------------------------------------------------------------------------
# SECTION 6: SAVE ALL CLEANED DATASETS
# --------------------------------------------------------------------------

write_csv(retail_clean,    "retail_clean.csv")
write_csv(prospect_clean,  "prospect_clean.csv")
write_csv(conjoint_clean,  "conjoint_clean.csv")
write_csv(profiles_clean,  "profiles_clean.csv")
write_csv(prod_attr_clean, "prod_attr_clean.csv")

cat("\n=== All Cleaned Datasets Saved Successfully ===\n")
cat("retail_clean.csv    :", nrow(retail_clean),    "rows,", ncol(retail_clean),
"cols\n")
cat("prospect_clean.csv   :", nrow(prospect_clean),   "rows,",
ncol(prospect_clean),  "cols\n")
cat("conjoint_clean.csv   :", nrow(conjoint_clean),  "rows,", ncol(conjoint_clean),
"cols\n")
cat("profiles_clean.csv    :", nrow(profiles_clean),   "rows,", ncol(profiles_clean),
"cols\n")
cat("prod_attr_clean.csv   :", nrow(prod_attr_clean),  "rows,",
ncol(prod_attr_clean), "cols\n")

#
============================================================
===========================
# PART 1 — STEP 2: RFM ANALYSIS
# PART 1 — STEP 3: CLUSTER ANALYSIS (Hierarchical + K-Means)
#
============================================================
===========================

# I loaded the cleaned retail dataset produced in Step 1.
retail_clean <- read_csv("C:/Users/Jai Baloda/Desktop/Jai DELL/Masters Business
Analytics/Semester 2/Marketing Analytics/retail_clean.csv", show_col_types =
FALSE) %>%
  mutate(
    InvoiceDate = as.Date(InvoiceDate),
```

```
    Married     = factor(Married, levels = c("Not Married", "Married")),
    Education   = factor(Education, levels = c("No University Degree",
                                               "Undergraduate", "Postgraduate")),
    Work        = factor(Work, levels = c("Health Services", "Financial Services",
                                          "Sales", "Advertising / PR", "Education",
                                          "Construction / Logistics", "Engineering",
                                          "Technology", "Retailing / Services",
                                          "SME / Self-Employed", "Transportation"))
  )

cat("Retail clean loaded:", nrow(retail_clean), "rows,",
    n_distinct(retail_clean$CustomerID), "unique customers\n")

# --------------------------------------------------------------------------
# SECTION 7: CALCULATE RECENCY, FREQUENCY, AND MONETARY (RFM)
# --------------------------------------------------------------------------

# I calculated the three RFM dimensions for each unique customer.
# The reference date was set to one day after the most recent transaction
# (2025-08-17), ensuring the most recent customer receives a Recency of 1 day.

ref_date <- max(retail_clean$InvoiceDate) + 1
cat("Reference date for Recency calculation:", as.character(ref_date), "\n")

rfm_raw <- retail_clean %>%
  group_by(CustomerID) %>%
  summarise(
    Recency   = as.numeric(ref_date - max(InvoiceDate)),
    Frequency = n_distinct(InvoiceNo),
    Monetary  = sum(TotalSpend),
    .groups   = "drop"
  )
```

```r
cat("\nRFM table dimensions:", dim(rfm_raw), "\n")
cat("\n--- RFM Descriptive Statistics ---\n")
summary(rfm_raw[, c("Recency", "Frequency", "Monetary")])

# --------------------------------------------------------------------------
# SECTION 8: RFM SCORING — SEQUENTIAL SORTING METHOD
# --------------------------------------------------------------------------

# I applied the sequential sorting method to assign RFM scores. Customers are
# first ranked by Recency, then Frequency within Recency groups, then Monetary
# within Recency-Frequency groups. This hierarchical approach reflects the
# convention where Recency is the strongest predictor of future behaviour
# (Palmatier et al., 2022).

rfm_sequential <- rfm_raw %>%
  mutate(R_score = ntile(-Recency, 5)) %>%
  group_by(R_score) %>%
  mutate(F_score = ntile(Frequency, 5)) %>%
  group_by(R_score, F_score) %>%
  mutate(M_score = ntile(Monetary, 5)) %>%
  ungroup() %>%
  mutate(
    RFM_Score_Sequential = R_score + F_score + M_score,
    RFM_Cell_Sequential  = paste0(R_score, F_score, M_score)
  )

cat("\n--- Sequential RFM Score Distribution ---\n")
print(table(rfm_sequential$RFM_Score_Sequential))

# I assigned customer segment labels based on the sequential RFM total score.
rfm_sequential <- rfm_sequential %>%
  mutate(Segment_Sequential = case_when(
```

```
    RFM_Score_Sequential >= 13 ~ "Champions",
    RFM_Score_Sequential >= 10 ~ "Loyal Customers",
    RFM_Score_Sequential >= 8  ~ "Potential Loyalists",
    RFM_Score_Sequential >= 5  ~ "At Risk",
    TRUE                       ~ "Lost Customers"
  )) %>%
  mutate(Segment_Sequential = factor(Segment_Sequential,
                                     levels = c("Champions", "Loyal Customers",
                                                "Potential Loyalists", "At Risk",
                                                "Lost Customers")))

cat("\n--- Sequential Segment Distribution ---\n")
print(table(rfm_sequential$Segment_Sequential))

# --------------------------------------------------------------------------
# SECTION 9: RFM SCORING — INDEPENDENT SORTING METHOD
# --------------------------------------------------------------------------

# I applied the independent sorting method where Recency, Frequency, and
# Monetary are each scored independently using quintiles, without nesting.

rfm_independent <- rfm_raw %>%
  mutate(
    R_score = ntile(-Recency,  5),
    F_score = ntile(Frequency, 5),
    M_score = ntile(Monetary,  5)
  ) %>%
  mutate(
    RFM_Score_Independent = R_score + F_score + M_score,
    RFM_Cell_Independent  = paste0(R_score, F_score, M_score)
  )

cat("\n--- Independent RFM Score Distribution ---\n")
```

```
print(table(rfm_independent$RFM_Score_Independent))

rfm_independent <- rfm_independent %>%
  mutate(Segment_Independent = case_when(
    RFM_Score_Independent >= 13 ~ "Champions",
    RFM_Score_Independent >= 10 ~ "Loyal Customers",
    RFM_Score_Independent >= 8  ~ "Potential Loyalists",
    RFM_Score_Independent >= 5  ~ "At Risk",
    TRUE                        ~ "Lost Customers"
  )) %>%
  mutate(Segment_Independent = factor(Segment_Independent,
                                      levels = c("Champions", "Loyal Customers",
                                                 "Potential Loyalists", "At Risk",
                                                 "Lost Customers")))

cat("\n--- Independent Segment Distribution ---\n")
print(table(rfm_independent$Segment_Independent))

# ---------------------------------------------------------------------------
# SECTION 10: RFM TABLES
# ---------------------------------------------------------------------------

cat("\n=== RFM TABLE — SEQUENTIAL METHOD ===\n")
rfm_table_sequential <- rfm_sequential %>%
  group_by(Segment_Sequential) %>%
  summarise(Customers = n(), Avg_Recency = round(mean(Recency), 1),
            Avg_Frequency = round(mean(Frequency), 2),
            Avg_Monetary  = round(mean(Monetary), 2),
            Total_Revenue = round(sum(Monetary), 2), .groups = "drop") %>%
  arrange(desc(Avg_Monetary))
print(rfm_table_sequential)

cat("\n=== RFM TABLE — INDEPENDENT METHOD ===\n")
```

```
rfm_table_independent <- rfm_independent %>%
  group_by(Segment_Independent) %>%
  summarise(Customers = n(), Avg_Recency = round(mean(Recency), 1),
            Avg_Frequency = round(mean(Frequency), 2),
            Avg_Monetary  = round(mean(Monetary), 2),
            Total_Revenue = round(sum(Monetary), 2), .groups = "drop") %>%
  arrange(desc(Avg_Monetary))
print(rfm_table_independent)

cat("\n=== METHOD COMPARISON: Segment Assignment Differences ===\n")
comparison <- rfm_sequential %>%
  select(CustomerID, Segment_Sequential) %>%
  left_join(rfm_independent %>% select(CustomerID, Segment_Independent),
            by = "CustomerID") %>%
  mutate(Agreement = Segment_Sequential == Segment_Independent)

cat("Customers assigned to same segment by both methods:",
    sum(comparison$Agreement), "/", nrow(comparison),
    paste0("(", round(mean(comparison$Agreement)*100, 1), "%)\n"))

rfm_combined <- rfm_sequential %>%
  left_join(
    rfm_independent %>%
      select(CustomerID, R_score_ind = R_score, F_score_ind = F_score,
             M_score_ind = M_score, RFM_Score_Independent,
             RFM_Cell_Independent, Segment_Independent),
    by = "CustomerID"
  )

# ---------------------------------------------------------------------------
# SECTION 11: RFM VISUALISATIONS (Figures 9–10)
# ---------------------------------------------------------------------------
```

```
seg_colours <- c(
  "Champions"           = "#2E86AB",
  "Loyal Customers"     = "#3A7D44",
  "Potential Loyalists" = "#F4A261",
  "At Risk"             = "#E07B39",
  "Lost Customers"      = "#C0392B"
)

# ---- FIGURE 9: RFM Segment Comparison — Sequential vs. Independent ----
seg_seq <- rfm_sequential %>% count(Segment_Sequential) %>%
  rename(Segment = Segment_Sequential) %>% mutate(Method = "Sequential")
seg_ind <- rfm_independent %>% count(Segment_Independent) %>%
  rename(Segment = Segment_Independent) %>% mutate(Method = "Independent")

seg_compare <- bind_rows(seg_seq, seg_ind) %>%
  mutate(Method = factor(Method, levels = c("Sequential", "Independent")))

p9 <- ggplot(seg_compare, aes(x = Segment, y = n, fill = Segment)) +
  geom_col(alpha = 0.85, show.legend = FALSE) +
  geom_text(aes(label = n), vjust = -0.4, size = 3.2, fontface = "bold") +
  facet_wrap(~ Method) +
  scale_fill_manual(values = seg_colours) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.14))) +
  labs(
    title    = "Figure 9: RFM Segment Distribution — Sequential vs. Independent Method",
    subtitle = "Number of customers assigned to each segment under both scoring methods",
    x = "Customer Segment", y = "Number of Customers"
  ) +
  theme_assignment() +
  theme(axis.text.x = element_text(angle = 25, hjust = 1))
print(p9)
```

```
# ---- FIGURE 10: RFM Segment Profiles ----
rfm_profile <- rfm_independent %>%
  group_by(Segment_Independent) %>%
  summarise(Recency = mean(Recency), Frequency = mean(Frequency),
            Monetary = mean(Monetary), .groups = "drop") %>%
  pivot_longer(-Segment_Independent, names_to = "Metric", values_to = "Value")
%>%
  mutate(
    Metric = factor(Metric, levels = c("Recency", "Frequency", "Monetary")),
    Segment_Independent = factor(Segment_Independent,
                             levels = c("Champions", "Loyal Customers",
                                        "Potential Loyalists", "At Risk",
                                        "Lost Customers"))
  )

p10 <- ggplot(rfm_profile,
           aes(x = Segment_Independent, y = Value, fill = Segment_Independent)) +
  geom_col(alpha = 0.85, show.legend = FALSE) +
  geom_text(aes(label = round(Value, 1)), vjust = -0.3, size = 3.0) +
  facet_wrap(~ Metric, scales = "free_y") +
  scale_fill_manual(values = seg_colours) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.16))) +
  labs(
    title    = "Figure 10: RFM Segment Profiles (Independent Method)",
     subtitle = "Mean Recency (days), Frequency (orders), and Monetary (£) per
segment",
    x = "Customer Segment", y = "Mean Value"
  ) +
  theme_assignment() +
  theme(axis.text.x = element_text(angle = 25, hjust = 1))
print(p10)
```

```
# --------------------------------------------------------------------------
# SECTION 12: DATA PREPARATION FOR CLUSTERING
# --------------------------------------------------------------------------

# I standardised RFM data to zero mean and unit variance before clustering.
# Without standardisation, unscaled Monetary values would dominate distances.

rfm_scaled <- rfm_raw %>%
  column_to_rownames("CustomerID") %>%
  scale() %>%
  as.data.frame()

cat("\n--- Scaled RFM Summary (should have mean ~0, sd ~1) ---\n")
print(summary(rfm_scaled))

# --------------------------------------------------------------------------
# SECTION 13: HIERARCHICAL CLUSTERING
# --------------------------------------------------------------------------

# I performed hierarchical clustering using Ward's minimum variance linkage,
# which minimises total within-cluster variance at each merge step and produces
# compact, well-separated clusters (Palmatier et al., 2022).

dist_matrix <- dist(rfm_scaled, method = "euclidean")
hc_model    <- hclust(dist_matrix, method = "ward.D2")

cat("\nHierarchical clustering completed using Ward's linkage.\n")
cat("Number of observations:", length(hc_model$order), "\n")

# ---- FIGURE 11: Dendrogram ----
dend          <- as.dendrogram(hc_model)
```

```
dend_coloured <- color_branches(dend, k = 4, col = brewer.pal(4, "Set1"))

p11_func <- function() {
  par(mar = c(4, 4, 4, 2))
  plot(dend_coloured,
       main = "Figure 11: Dendrogram — Hierarchical Clustering (Ward's Method)",
       sub  = "Cut at k = 4 clusters; branch colours indicate cluster assignment",
       ylab = "Height (Within-cluster variance)", leaflab = "none",
       cex.main = 1.1, cex.sub = 0.85, cex.lab = 0.95)
  abline(h = 10, col = "red", lty = 2, lwd = 1.5)
  legend("topright", legend = paste("Cluster", 1:4),
         fill = brewer.pal(4, "Set1"), bty = "n", cex = 0.85)
}
p11_func()

hc_clusters <- cutree(hc_model, k = 4)
cat("\nHierarchical cluster sizes (k=4):\n")
print(table(hc_clusters))

# ---------------------------------------------------------------------------
# SECTION 14: ELBOW METHOD — OPTIMAL NUMBER OF CLUSTERS
# ---------------------------------------------------------------------------

# I applied the elbow method to identify the optimal k for K-means clustering.
# The optimal k is the point where additional clusters yield diminishing
# returns in WSS reduction (James et al., 2013).

set.seed(40485958)
wss_values <- map_dbl(1:10, function(k) {
  km <- kmeans(rfm_scaled, centers = k, nstart = 25, iter.max = 300)
  km$tot.withinss
})
```

```
elbow_df <- tibble(k = 1:10, WSS = wss_values)
cat("\n--- Within-cluster Sum of Squares by k ---\n")
print(elbow_df)

# ---- FIGURE 12: Elbow Method Plot ----
p12 <- ggplot(elbow_df, aes(x = k, y = WSS)) +
  geom_line(colour = "#2E86AB", linewidth = 1.3) +
  geom_point(colour = "#2E86AB", size = 3.5) +
  geom_point(data = filter(elbow_df, k == 4), aes(x = k, y = WSS),
             colour = "#C0392B", size = 5, shape = 18) +
  annotate("text", x = 4.3, y = wss_values[4] + 15,
           label = "Optimal k = 4\n(elbow point)", colour = "#C0392B",
           size = 3.5, hjust = 0) +
  geom_vline(xintercept = 4, colour = "#C0392B", linetype = "dashed", linewidth
= 0.8) +
  scale_x_continuous(breaks = 1:10) +
  labs(
    title    = "Figure 12: Elbow Method — Optimal Number of K-Means Clusters",
    subtitle = "Total within-cluster sum of squares (WSS) for k = 1 to 10",
    x = "Number of Clusters (k)", y = "Total Within-Cluster SS (WSS)"
  ) +
  theme_assignment()
print(p12)

# ---------------------------------------------------------------------------
# SECTION 15: K-MEANS CLUSTERING WITH DIFFERENT INITIAL START POINTS
# ---------------------------------------------------------------------------

# I applied K-means with three different random seed configurations to address
# sensitivity to initial centroid placement. Each run uses nstart = 25.
# The solution with the lowest total WSS is retained as the final model
# (James et al., 2013).
```

```
set.seed(42)

km_run1 <- kmeans(rfm_scaled, centers = 4, nstart = 25, iter.max = 300)

cat("\nK-Means Run 1 (seed=42): Total WSS =", round(km_run1$tot.withinss, 2),

    "| Iterations =", km_run1$iter, "\n")

cat("Cluster sizes:", km_run1$size, "\n")


set.seed(123)

km_run2 <- kmeans(rfm_scaled, centers = 4, nstart = 25, iter.max = 300)

cat("\nK-Means Run 2 (seed=123): Total WSS =", round(km_run2$tot.withinss, 2),

    "| Iterations =", km_run2$iter, "\n")

cat("Cluster sizes:", km_run2$size, "\n")


set.seed(2025)

km_run3 <- kmeans(rfm_scaled, centers = 4, nstart = 25, iter.max = 300)

cat("\nK-Means Run 3 (seed=2025): Total WSS =", round(km_run3$tot.withinss, 2),

    "| Iterations =", km_run3$iter, "\n")

cat("Cluster sizes:", km_run3$size, "\n")


best_wss <- min(km_run1$tot.withinss, km_run2$tot.withinss, km_run3$tot.withinss)

km_final <- if (km_run1$tot.withinss == best_wss) km_run1 else

  if (km_run2$tot.withinss == best_wss) km_run2 else km_run3


cat("\nFinal K-Means model selected: Total WSS =", round(km_final$tot.withinss, 2), "\n")

cat("Final cluster sizes:", km_final$size, "\n")


comparison_df <- tibble(

  Run   = c("Run 1 (seed=42)", "Run 2 (seed=123)", "Run 3 (seed=2025)"),

  WSS   = c(km_run1$tot.withinss, km_run2$tot.withinss, km_run3$tot.withinss),

  Iters = c(km_run1$iter, km_run2$iter, km_run3$iter)

)
```

```r
cat("\n--- K-Means Stability Comparison ---\n")
print(comparison_df)

# -----------------------------------------------------------------------
# SECTION 16: CLUSTER ASSIGNMENT AND LABELLING
# -----------------------------------------------------------------------

rfm_clustered <- rfm_raw %>%
  mutate(KMeans_Cluster = factor(km_final$cluster))

rfm_clustered <- rfm_clustered %>%
  left_join(
    rfm_independent %>%
      select(CustomerID, R_score, F_score, M_score,
             RFM_Score_Independent, Segment_Independent),
    by = "CustomerID"
  )

cat("\n=== K-Means Cluster Centroids (Original Units) ===\n")
cluster_profiles <- rfm_clustered %>%
  group_by(KMeans_Cluster) %>%
  summarise(n_customers   = n(),
            Avg_Recency   = round(mean(Recency),   1),
            Avg_Frequency = round(mean(Frequency), 2),
            Avg_Monetary  = round(mean(Monetary),  2), .groups = "drop")
print(cluster_profiles)

# I labelled clusters by inspecting centroid characteristics:
# highest monetary + lowest recency = Champions; lowest across all = Lost Customers.
cluster_labels <- cluster_profiles %>%
  arrange(desc(Avg_Monetary)) %>%
  mutate(Cluster_Label  = c("Champions", "Loyal Customers",
```

```
                    "At Risk", "Lost Customers"),
       KMeans_Cluster = as.character(KMeans_Cluster)) %>%
  select(KMeans_Cluster, Cluster_Label)

# I dropped any existing Cluster_Label column before joining to prevent
# duplicate column conflicts if this block is re-run.
rfm_clustered <- rfm_clustered %>%
  mutate(KMeans_Cluster = as.character(KMeans_Cluster)) %>%
  select(-any_of("Cluster_Label")) %>%
  left_join(cluster_labels, by = "KMeans_Cluster") %>%
  mutate(KMeans_Cluster = factor(KMeans_Cluster),
       Cluster_Label  = factor(Cluster_Label,
                          levels = c("Champions", "Loyal Customers",
                                 "At Risk", "Lost Customers")))

cat("\n--- Final Cluster Label Assignment ---\n")
print(rfm_clustered %>% count(KMeans_Cluster, Cluster_Label) %>%
arrange(KMeans_Cluster))

cluster_colours <- c(
  "Champions"      = "#2E86AB",
  "Loyal Customers" = "#3A7D44",
  "At Risk"        = "#F4A261",
  "Lost Customers"  = "#C0392B"
)

# ---- FIGURE 13: K-Means Cluster Scatter Plot ----
p13 <- ggplot(rfm_clustered, aes(x = Recency, y = Monetary, colour =
Cluster_Label)) +
  geom_point(alpha = 0.55, size = 1.8) +
  stat_ellipse(aes(fill = Cluster_Label), type = "norm", level = 0.75,
           geom = "polygon", alpha = 0.08, show.legend = FALSE) +
  scale_colour_manual(values = cluster_colours) +
```

```
  scale_fill_manual(values   = cluster_colours) +
  scale_y_continuous(labels  = label_dollar(prefix = "£")) +
  labs(
    title    = "Figure 13: K-Means Cluster Solution — Recency vs. Monetary",
    subtitle = "k = 4 clusters; ellipses show 75% confidence region per cluster",
   x = "Recency (days since last purchase)", y = "Total Monetary Value (£)", colour
= "Cluster"
  ) +
  theme_assignment()
print(p13)

# -----------------------------------------------------------------------
# SECTION 17: SAVE CLUSTERING OUTPUTS
# -----------------------------------------------------------------------

write_csv(rfm_clustered,         "rfm_clustered.csv")
write_csv(rfm_sequential,        "rfm_sequential.csv")
write_csv(rfm_independent,       "rfm_independent.csv")
write_csv(rfm_table_sequential,  "rfm_table_sequential.csv")
write_csv(rfm_table_independent, "rfm_table_independent.csv")

cat("\n=== All RFM and Clustering outputs saved ===\n")
cat("rfm_clustered.csv:", nrow(rfm_clustered), "customers,",
    n_distinct(rfm_clustered$KMeans_Cluster), "clusters\n")

#
=============================================================
======================
# PART 1 — STEP 4: CUSTOMER PROFILING
# PART 1 — STEP 5: LDA, ANOVA, AND CONFUSION MATRIX
# PART 1 — STEP 6: TARGET SEGMENT SELECTION AND JUSTIFICATION
#
=============================================================
======================
```

```
# I loaded the cleaned retail dataset and the clustered RFM output.

retail_clean <- read_csv("C:/Users/Jai Baloda/Desktop/Jai DELL/Masters Business Analytics/Semester 2/Marketing Analytics/retail_clean.csv", show_col_types = FALSE) %>%
  mutate(InvoiceDate = as.Date(InvoiceDate))

rfm_clustered <- read_csv("C:/Users/Jai Baloda/Desktop/Jai DELL/Masters Business Analytics/Semester 2/Marketing Analytics/rfm_clustered.csv", show_col_types = FALSE)

prospect_clean <- read_csv("C:/Users/Jai Baloda/Desktop/Jai DELL/Masters Business Analytics/Semester 2/Marketing Analytics/prospect_clean.csv", show_col_types = FALSE)


# --------------------------------------------------------------------------
# SECTION 18: MERGE CLUSTER ASSIGNMENTS WITH DEMOGRAPHIC DATA
# --------------------------------------------------------------------------


# I extracted one demographic record per customer and merged with clustered RFM.


demographics <- retail_clean %>%
  distinct(CustomerID, .keep_all = TRUE) %>%
  select(CustomerID, Age, Income, HouseholdSize, Married, Education, Work)


rfm_full <- rfm_clustered %>%
  left_join(demographics, by = "CustomerID") %>%
  mutate(
    Married   = factor(Married, levels = c("Not Married", "Married")),
    Education = factor(Education, levels = c("No University Degree",
                                             "Undergraduate", "Postgraduate")),
    Work      = factor(Work, levels = c("Health Services", "Financial Services",
                                        "Sales", "Advertising / PR", "Education",
                                        "Construction / Logistics", "Engineering",
                                        "Technology", "Retailing / Services",
                                        "SME / Self-Employed", "Transportation"))
  )
```

```
cat("RFM + Demographics dataset:", nrow(rfm_full), "customers\n")
cat("Cluster breakdown:\n")
print(table(rfm_full$Cluster_Label))

# --------------------------------------------------------------------------
# SECTION 19: CLUSTER PROFILE SUMMARY TABLE
# --------------------------------------------------------------------------

cluster_profile_table <- rfm_full %>%
  group_by(Cluster_Label) %>%
  summarise(
    N_Customers  = n(),
    Avg_Recency  = round(mean(Recency),      1),
    Avg_Frequency= round(mean(Frequency),    2),
    Avg_Monetary = round(mean(Monetary),     2),
    Avg_Age      = round(mean(Age),          1),
    Avg_Income_k = round(mean(Income),       1),
    Avg_HH_Size  = round(mean(HouseholdSize),2),
    Pct_Married  = round(mean(Married == "Married") * 100, 1),
    .groups      = "drop"
  ) %>%
  arrange(desc(Avg_Monetary))

cat("\n=== CUSTOMER PROFILE TABLE — K-MEANS CLUSTERS (k=4) ===\n")
print(cluster_profile_table)

cat("\n=== Education Distribution by Cluster ===\n")
edu_by_cluster <- rfm_full %>%
  group_by(Cluster_Label, Education) %>% summarise(n = n(), .groups = "drop")
%>%
   group_by(Cluster_Label) %>% mutate(Pct = round(n / sum(n) * 100, 1)) %>%
ungroup()
```

```
print(edu_by_cluster)

cat("\n=== Occupation Distribution by Cluster ===\n")
work_by_cluster <- rfm_full %>%
  group_by(Cluster_Label, Work) %>% summarise(n = n(), .groups = "drop") %>%
  group_by(Cluster_Label) %>% mutate(Pct = round(n / sum(n) * 100, 1)) %>%
  arrange(Cluster_Label, desc(Pct)) %>% ungroup()
print(work_by_cluster)

# Cluster profile narrative:
# CHAMPIONS (n=51):  Recency 18d, Frequency 12.2, Monetary £362 — most engaged,
#                    highest-value customers.
# LOYAL CUSTOMERS (n=220): Recency 32d, Frequency 5.7, Monetary £147 — consistent
#                    engagement, clear growth pathway to Champions.
# AT RISK (n=414):   Recency 48d, Frequency 2.1, Monetary £45 — declining
#                    engagement, at risk of lapsing.
# LOST CUSTOMERS (n=237): Recency 142d, Frequency 1.4, Monetary £33 — highly
#                    disengaged, reactivation requires significant investment.

# ---------------------------------------------------------------------------
# SECTION 20: CUSTOMER PROFILING VISUALISATIONS (Figures 14–15)
# ---------------------------------------------------------------------------

# ---- FIGURE 14: Radar Chart — Cluster Profiles ----
profile_radar <- cluster_profile_table %>%
  select(Cluster_Label, Avg_Recency, Avg_Frequency,
         Avg_Monetary, Avg_Age, Avg_Income_k) %>%
  mutate(Avg_Recency   = 1 - rescale(Avg_Recency),
         Avg_Frequency = rescale(Avg_Frequency),
         Avg_Monetary  = rescale(Avg_Monetary),
         Avg_Age       = rescale(Avg_Age),
```

```
        Avg_Income_k  = rescale(Avg_Income_k)) %>%
  column_to_rownames("Cluster_Label")

colnames(profile_radar) <- c("Recency\n(inversed)", "Frequency",
                             "Monetary", "Age", "Income")

radar_data    <- rbind(rep(1, 5), rep(0, 5), profile_radar)
radar_colours <- c("#2E86AB", "#3A7D44", "#F4A261", "#C0392B")

p14_func <- function() {
  par(mar = c(1, 1, 2, 1))
  radarchart(radar_data, axistype = 1, pcol = radar_colours,
             pfcol = adjustcolor(radar_colours, alpha.f = 0.15), plwd = 2.5,
             cglcol = "grey70", cglty = 1, axislabcol = "grey40",
             caxislabels = c("0","0.25","0.5","0.75","1"), vlcex = 0.9,
            title = "Figure 14: Customer Cluster Profiles — Radar Chart\n(Normalised dimensions)")
  legend(x = "topright", legend = rownames(profile_radar),
         col = radar_colours, lty = 1, lwd = 2, bty = "n", cex = 0.82)
}
p14_func()

# ---- FIGURE 15: Cluster Size and Revenue Contribution ----
cluster_revenue <- rfm_full %>%
  group_by(Cluster_Label) %>%
   summarise(N_Customers = n(), Total_Revenue = sum(Monetary), .groups = "drop") %>%
  mutate(Revenue_Share = Total_Revenue / sum(Total_Revenue) * 100)

p15a <- ggplot(cluster_revenue,
               aes(x = reorder(Cluster_Label, -N_Customers),
                   y = N_Customers, fill = Cluster_Label)) +
  geom_col(alpha = 0.85, show.legend = FALSE) +
```

```
 geom_text(aes(label = N_Customers), vjust = -0.4, size = 3.8, fontface = "bold")
+
  scale_fill_manual(values = cluster_colours) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.14))) +
  labs(title = "Customer Count per Cluster", x = "Cluster", y = "Number of
Customers") +
  theme_assignment()

p15b <- ggplot(cluster_revenue,
               aes(x = reorder(Cluster_Label, -Total_Revenue),
                   y = Total_Revenue, fill = Cluster_Label)) +
  geom_col(alpha = 0.85, show.legend = FALSE) +
  geom_text(aes(label = paste0("£", round(Total_Revenue/1000, 1), "k\n(",
                               round(Revenue_Share, 1), "%)")),
            vjust = -0.3, size = 3.2, lineheight = 1.1) +
  scale_fill_manual(values = cluster_colours) +
  scale_y_continuous(labels = label_dollar(prefix = "£"),
                     expand = expansion(mult = c(0, 0.18))) +
  labs(title = "Total Revenue Contribution per Cluster",
       x = "Cluster", y = "Total Revenue (£)") +
  theme_assignment()

p15 <- p15a + p15b +
  plot_annotation(
    title    = "Figure 15: Cluster Size and Revenue Contribution",
    subtitle = "Customer count and cumulative monetary value generated per
segment",
    theme    = theme(plot.title    = element_text(face = "bold", size = 13, hjust =
0.5),
                     plot.subtitle = element_text(size = 10, hjust = 0.5, colour = "grey45")))
print(p15)

# ---------------------------------------------------------------------------
# SECTION 21: LDA MODEL — TRAINING AND EVALUATION
```

```
# ---------------------------------------------------------------------------

# I applied LDA to assess cluster separability and to classify prospects.
# The model uses both RFM (behavioural) and demographic features for maximum
# discriminant power (Palmatier et al., 2022).

rfm_lda_data <- rfm_full %>%
  mutate(Married_num   = as.numeric(Married) - 1,
         Education_num = as.numeric(Education),
         Work_num      = as.numeric(Work),
         Cluster_num   = as.numeric(KMeans_Cluster))

lda_features <- c("Recency", "Frequency", "Monetary",
                  "Age", "Income", "HouseholdSize",
                  "Married_num", "Education_num", "Work_num")

set.seed(45)
train_idx  <- createDataPartition(rfm_lda_data$KMeans_Cluster, p = 0.80, list = FALSE)
train_data <- rfm_lda_data[ train_idx, ]
test_data  <- rfm_lda_data[-train_idx, ]

cat("\nTraining set:", nrow(train_data), "customers\n")
cat("Test set    :", nrow(test_data),  "customers\n")
cat("Train cluster distribution:\n"); print(table(train_data$Cluster_Label))
cat("Test cluster distribution:\n");  print(table(test_data$Cluster_Label))

# I constructed the model formula and fitted LDA on the training split.
lda_formula <- as.formula(paste("KMeans_Cluster ~",
                                paste(lda_features, collapse = " + ")))
lda_model   <- lda(lda_formula, data = train_data)
```

```
cat("\n=== LDA MODEL SUMMARY ===\n")
print(lda_model)

# I generated class predictions on training and test sets.
pred_train_obj <- predict(lda_model, newdata = train_data)
pred_train     <- pred_train_obj$class
pred_test_obj  <- predict(lda_model, newdata = test_data)
pred_test      <- pred_test_obj$class

train_acc <- mean(pred_train == train_data$KMeans_Cluster)
test_acc  <- mean(pred_test  == test_data$KMeans_Cluster)

cat(sprintf("\nTraining Accuracy : %.2f%%\n", train_acc * 100))
cat(sprintf("Test Accuracy     : %.2f%%\n",  test_acc  * 100))

# ── ANOVA TEST ────────────────────────────────────────────
# I ran one-way ANOVA on each RFM feature to formally test whether cluster means
# are statistically different — a necessary condition for meaningful LDA.
cat("\n=== ANOVA: Cluster Separation Tests ===\n")
for (var in c("Recency", "Frequency", "Monetary")) {
  aov_result <- aov(as.formula(paste(var, "~ KMeans_Cluster")), data = rfm_lda_data)
  cat(sprintf("\n%s ~ Cluster:\n", var))
  print(summary(aov_result))
}

# ── CONFUSION MATRIX ────────────────────────────────────────
test_actual_factor <- factor(test_data$KMeans_Cluster)
test_pred_factor   <- factor(pred_test, levels = levels(test_actual_factor))

cm <- confusionMatrix(data = test_pred_factor, reference = test_actual_factor)
```

```
cat("\n=== CONFUSION MATRIX — TEST SET ===\n")
print(cm)

cm_df <- as.data.frame(cm$table) %>% rename(Predicted = Prediction, Actual = Reference)
level_labels <- c("1" = "At Risk", "2" = "Lost Customers",
                  "3" = "Champions", "4" = "Loyal Customers")

cm_df <- cm_df %>%
  mutate(Predicted = recode(as.character(Predicted), !!!level_labels),
         Actual    = recode(as.character(Actual),    !!!level_labels))

# ── PROSPECT PREDICTION ──────────────────────────────────────────

# I applied the fitted LDA model to all 5,000 prospect customers.
prospect_lda_data <- prospect_clean %>%
  mutate(
    Married_num   = as.numeric(factor(Married, levels = c("Not Married", "Married"))) - 1,
    Education_num = as.numeric(factor(Education, levels = c("No University Degree",
                                                            "Undergraduate",
                                                            "Postgraduate"))),
    Work_num      = as.numeric(factor(Work,
                               levels = c("Health Services", "Financial Services",
                                          "Sales", "Advertising / PR", "Education",
                                          "Construction / Logistics", "Engineering",
                                          "Technology", "Retailing / Services",
                                          "SME / Self-Employed", "Transportation")))
  )

# I created dummy RFM columns set to training medians, since prospects have
# no transaction history. Demographics drive the prediction in practice.
prospect_lda_data <- prospect_lda_data %>%
```

```
  mutate(Recency   = median(train_data$Recency),
         Frequency = median(train_data$Frequency),
         Monetary  = median(train_data$Monetary))

prospect_pred_obj <- predict(lda_model, newdata = prospect_lda_data)
prospect_clean    <- prospect_clean %>%
  mutate(KMeans_Cluster = prospect_pred_obj$class,
         Cluster_Label  = recode(as.character(KMeans_Cluster), !!!level_labels),
         Cluster_Label  = factor(Cluster_Label,
                         levels = c("Champions", "Loyal Customers",
                                    "At Risk", "Lost Customers")))

cat("\n=== PROSPECT SEGMENT PREDICTIONS ===\n")
print(table(prospect_clean$Cluster_Label))

# ---- FIGURE 16: Confusion Matrix Heatmap + Prospect Distribution ----
p16a <- ggplot(cm_df, aes(x = Predicted, y = Actual, fill = Freq)) +
  geom_tile(colour = "white", linewidth = 0.8) +
  geom_text(aes(label = Freq), size = 5, fontface = "bold",
          colour = ifelse(cm_df$Freq > (max(cm_df$Freq) / 2), "white", "grey20")) +
  scale_fill_gradient(low = "#EAF4FB", high = "#2E86AB", name = "Count") +
  labs(title    = "Confusion Matrix (Test Set)",
       subtitle = paste0("Overall accuracy: ", round(test_acc * 100, 1), "%"),
       x = "Predicted Cluster", y = "Actual Cluster") +
  theme_assignment() +
   theme(axis.text.x = element_text(angle = 20, hjust = 1), legend.position = "right")

prospect_dist <- prospect_clean %>% count(Cluster_Label) %>% mutate(Pct = n / sum(n) * 100)

p16b <- ggplot(prospect_dist, aes(x = reorder(Cluster_Label, -n), y = n, fill = Cluster_Label)) +
  geom_col(alpha = 0.85, show.legend = FALSE) +
```

```
  geom_text(aes(label = paste0(n, "\n(", round(Pct, 1), "%)")),
            vjust = -0.3, size = 3.5, lineheight = 1.1) +
  scale_fill_manual(values = cluster_colours) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.18))) +
  labs(title    = "Prospect Customers: Predicted Segment",
       subtitle = "LDA-predicted cluster for 5,000 prospect customers",
       x = "Predicted Segment", y = "Number of Prospects") +
  theme_assignment()

p16 <- p16a + p16b +
  plot_annotation(
       title      = "Figure 16: LDA Performance — Confusion Matrix & Prospect Targeting",
      subtitle = "Left: test set classification accuracy. Right: prospect segment predictions.",
    theme    = theme(plot.title    = element_text(face = "bold", size = 13, hjust = 0.5),
                plot.subtitle = element_text(size = 10, hjust = 0.5, colour = "grey45")))
print(p16)

# ---------------------------------------------------------------------------
# SECTION 22: TARGET SEGMENT SELECTION AND JUSTIFICATION
# ---------------------------------------------------------------------------

# I selected LOYAL CUSTOMERS as the primary target segment for the new PB
# coffee product based on the following evidence:
#
# 1. SIZE: n=220 — commercially viable, focused enough for targeted strategy.
# 2. REVENUE: avg £147.24/customer, second-highest revenue pool (£32,392 total).
# 3. ENGAGEMENT: avg 32 days recency, 5.72 orders — active, regular purchasers.
# 4. GROWTH: clear pathway to Champions; bridging gap = strategic opportunity.
```

```
# 5. NOT CHAMPIONS: already maximally engaged; cannibalisation risk outweighs gain.

# 6. NOT AT RISK/LOST: insufficient frequency/spend to justify launch investment.

# 7. DEMOGRAPHICS: avg age 54, income £68k, HH size 2.84 — quality-conscious

#    adults aligned with a premium, sustainable PB coffee proposition.


cat("\n=== STEP 6: TARGET SEGMENT — LOYAL CUSTOMERS ===\n")

cat("\nKey metrics for selected target segment:\n")


loyal_profile <- rfm_full %>%

  filter(Cluster_Label == "Loyal Customers") %>%

  summarise(N_Customers   = n(),

            Avg_Recency   = round(mean(Recency),   1),

            Avg_Frequency = round(mean(Frequency), 2),

            Avg_Monetary  = round(mean(Monetary),  2),

            Total_Revenue = round(sum(Monetary),   2),

            Avg_Age       = round(mean(Age),       1),

            Avg_Income_k  = round(mean(Income),    1),

            Pct_Married   = round(mean(Married == "Married") * 100, 1),

            Avg_HH_Size   = round(mean(HouseholdSize), 2))

print(t(loyal_profile))


# ---- FIGURE 17: Target Segment Spotlight ----

spotlight_data <- rfm_full %>%

  group_by(Cluster_Label) %>%

    summarise(`Avg Monetary (£)` = mean(Monetary), `Avg Frequency` = mean(Frequency),

            `Avg Recency (days, lower=better)` = mean(Recency), .groups = "drop") %>%

  pivot_longer(-Cluster_Label, names_to = "Metric", values_to = "Value") %>%

  mutate(IsTarget      = ifelse(Cluster_Label == "Loyal Customers",

                              "Target Segment", "Other Segments"),

         Cluster_Label = factor(Cluster_Label,
```

```
                            levels = c("Champions", "Loyal Customers",
                                       "At Risk", "Lost Customers")))

p17 <- ggplot(spotlight_data,
          aes(x = Cluster_Label, y = Value, fill = Cluster_Label, alpha = IsTarget)) +
  geom_col() +
  geom_text(aes(label = round(Value, 1)), vjust = -0.4, size = 3.2) +
  facet_wrap(~ Metric, scales = "free_y") +
  scale_fill_manual(values = cluster_colours) +
  scale_alpha_manual(values = c("Target Segment" = 1.0, "Other Segments" = 0.45)) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.16))) +
  labs(title    = "Figure 17: Target Segment Selection — Loyal Customers Spotlight",
       subtitle = "Highlighted segment selected for PB coffee product targeting (Part 2)",
       x = "Customer Segment", y = "Mean Value", fill = "Cluster", alpha = "Role") +
  theme_assignment() +
  theme(axis.text.x = element_text(angle = 20, hjust = 1), legend.position = "bottom")
print(p17)

# ----------------------------------------------------------------------------
# SECTION 23: SAVE PART 1 OUTPUTS
# ----------------------------------------------------------------------------

write_csv(rfm_full,      "rfm_profiled.csv")
write_csv(prospect_clean, "prospect_targeted.csv")

cat("\n=== All outputs saved ===\n")
cat("rfm_profiled.csv      :", nrow(rfm_full),      "customers\n")
cat("prospect_targeted.csv :", nrow(prospect_clean), "prospects\n")
cat("\n=== PART 1 COMPLETE — Target segment: LOYAL CUSTOMERS (n=220) ===\n")
```

```
#
==============================================================================
# PART 2 — STEP 1: FRACTIONAL FACTORIAL DESIGN
# PART 2 — STEP 2: PART-WORTHS AND WILLINGNESS TO PAY
# PART 2 — STEP 3: ATTRIBUTE IMPORTANCE
#
==============================================================================

# I loaded the three Part 2 datasets.
conjoint_clean <- read_csv(
  "C:/Users/Jai Baloda/Desktop/Jai DELL/Masters Business Analytics/Semester 2/Marketing Analytics/Conjoint survey results-1.csv",
  show_col_types = FALSE) %>%
  mutate(format = factor(format, levels = c("Instant","Capsule","Ground","Whole bean")),
         strength = factor(strength, levels = c("Mild","Medium","Dark")),
         origin = factor(origin, levels = c("House blend","100% Arabica blend",
                                            "Single-origin")),
         sustainability = factor(sustainability, levels = c("No","Yes")))

profiles_clean <- read_csv(
  "C:/Users/Jai Baloda/Desktop/Jai DELL/Masters Business Analytics/Semester 2/Marketing Analytics/Product profiles.csv",
  show_col_types = FALSE) %>%
  mutate(format = factor(format, levels = c("Instant","Capsule","Ground","Whole bean")),
         strength = factor(strength, levels = c("Mild","Medium","Dark")),
         origin = factor(origin, levels = c("House blend","100% Arabica blend",
                                            "Single-origin")),
         sustainability = factor(sustainability, levels = c("No","Yes")))

prod_attr_clean <- read_csv(
  "C:/Users/Jai Baloda/Desktop/Jai DELL/Masters Business Analytics/Semester 2/Marketing Analytics/Product attributes information.csv",
```

```
  show_col_types = FALSE) %>%
  rename(sustainability_claim = sustaintability_claim) %>%
  mutate(brand = str_extract(product_name, "^[A-Za-z]+"))

cat("Conjoint loaded  :", nrow(conjoint_clean), "rows,",
    n_distinct(conjoint_clean$respondent_id), "respondents\n")
cat("Profiles loaded  :", nrow(profiles_clean), "profiles\n")
cat("Prod attr loaded :", nrow(prod_attr_clean), "products\n")

# -------------------------------------------------------------------------
# SECTION 24: STEP 1 — WHY ONLY 16 PRODUCT PROFILES?
# -------------------------------------------------------------------------

# I calculated the full factorial design size to demonstrate why a subset was
# required. With 5 attributes at 4, 4, 3, 3, and 2 levels, the total possible
# combinations are 4 x 4 x 3 x 3 x 2 = 288 profiles. Asking 200 respondents
# to rate all 288 is cognitively infeasible. A fractional factorial design
# selects the minimum orthogonal subset needed to estimate all part-worths.

full_factorial <- 4 * 4 * 3 * 3 * 2
cat("\n=== STEP 1: FRACTIONAL FACTORIAL DESIGN ===\n")
cat("Full factorial size (4 x 4 x 3 x 3 x 2):", full_factorial, "profiles\n")
cat("Selected orthogonal subset              :", nrow(profiles_clean), "profiles\n")
cat("Reduction factor                        :", round(full_factorial / nrow(profiles_clean), 1), "x\n")

cat("\n--- Attribute Level Frequency in the 16 Profiles ---\n")
cat("Price levels:\n");        print(table(profiles_clean$price))
cat("Format levels:\n");        print(table(profiles_clean$format))
cat("Strength levels:\n");      print(table(profiles_clean$strength))
cat("Origin levels:\n");        print(table(profiles_clean$origin))
cat("Sustainability levels:\n"); print(table(profiles_clean$sustainability))
```

```r
params_needed <- (4-1) + (4-1) + (3-1) + (3-1) + (2-1) + 1
df_residual   <- nrow(profiles_clean) - params_needed
cat("\nParameters to estimate (incl. intercept):", params_needed, "\n")
cat("Profiles available                       :", nrow(profiles_clean), "\n")
cat("Residual degrees of freedom              :", df_residual, "\n")
cat("Conclusion: 16 profiles is the minimum orthogonal subset that allows\n")
cat("all", params_needed - 1, "part-worths + intercept to be estimated independently.\n")


# ---------------------------------------------------------------------------
# SECTION 25: STEP 2 — PART-WORTH COMPUTATION AND WTP
# ---------------------------------------------------------------------------


# I computed individual-level part-worths by running a separate OLS regression
# for each of the 200 respondents. Reference levels: Price £3, Format Instant,
# Strength Mild, Origin House blend, Sustainability No.


cat("\n=== STEP 2: INDIVIDUAL-LEVEL PART-WORTHS ===\n")


respondent_ids <- unique(conjoint_clean$respondent_id)


partworth_list <- lapply(respondent_ids, function(rid) {
  df_r  <- conjoint_clean %>% filter(respondent_id == rid)
  model <- lm(rating ~ factor(price) + format + strength + origin + sustainability,
              data = df_r)
  coef_vec <- coef(model); coef_vec["respondent_id"] <- rid; coef_vec
})


all_coef_names <- unique(unlist(lapply(partworth_list, names)))
partworth_df <- do.call(rbind, lapply(partworth_list, function(x) {
  out <- setNames(rep(NA_real_, length(all_coef_names)), all_coef_names)
  out[names(x)] <- x; out
})) %>%
```

```r
  as.data.frame() %>%
  rename(Intercept       = `(Intercept)`,
         Price_4         = `factor(price)4`,
         Price_5         = `factor(price)5`,
         Price_6         = `factor(price)6`,
         Format_Capsule   = `formatCapsule`,
         Format_Ground    = `formatGround`,
         Format_Wholebean = `formatWhole bean`,
         Strength_Medium  = `strengthMedium`,
         Strength_Dark    = `strengthDark`,
         Origin_Arabica   = `origin100% Arabica blend`,
         Origin_Single    = `originSingle-origin`,
         Sustain_Yes      = `sustainabilityYes`) %>%
  mutate(respondent_id = respondent_ids)

cat("Part-worth matrix dimensions:", nrow(partworth_df), "respondents x",
    ncol(partworth_df) - 1, "coefficients\n")

mean_pw <- partworth_df %>%
  select(-respondent_id) %>%
  summarise(across(everything(), ~ mean(.x, na.rm = TRUE)))

cat("\n--- Mean Part-Worths Across 200 Respondents ---\n")
print(t(mean_pw))

# —— WILLINGNESS TO PAY (WTP) — AGGREGATE METHOD ——————————————

# I calculated WTP using aggregate mean part-worths — the standard approach
# (Orme, 2006; Palmatier et al., 2022). Individual-level price slopes can be
# near-zero for some respondents, causing extreme WTP values if averaged directly.

cat("\n=== WILLINGNESS TO PAY (WTP) — AGGREGATE METHOD ===\n")
```

```
mean_pw_price_vec <- c(0, mean_pw$Price_4, mean_pw$Price_5, mean_pw$Price_6)

price_vals_vec   <- c(3, 4, 5, 6)

agg_price_slope  <- coef(lm(mean_pw_price_vec ~ price_vals_vec))[2]


cat(sprintf("Aggregate price sensitivity: %.4f utility per £1 increase\n", agg_price_slope))


wtp_agg <- tibble(

  Attribute = c("Format: Capsule","Format: Ground","Format: Whole Bean",

                "Strength: Medium","Strength: Dark",

                "Origin: 100% Arabica","Origin: Single-Origin","Sustainability: Yes"),

  PartWorth = c(mean_pw$Format_Capsule, mean_pw$Format_Ground, mean_pw$Format_Wholebean,

                mean_pw$Strength_Medium, mean_pw$Strength_Dark,

                mean_pw$Origin_Arabica, mean_pw$Origin_Single, mean_pw$Sustain_Yes),

  WTP = c(-mean_pw$Format_Capsule, -mean_pw$Format_Ground, -mean_pw$Format_Wholebean,

          -mean_pw$Strength_Medium, -mean_pw$Strength_Dark,

          -mean_pw$Origin_Arabica, -mean_pw$Origin_Single,

          -mean_pw$Sustain_Yes) / agg_price_slope,

  Category = c("Format","Format","Format","Strength","Strength",

               "Origin","Origin","Sustainability")

) %>% arrange(desc(WTP))


cat("\n--- Aggregate WTP (£, vs reference levels) ---\n")

cat("Reference: Instant | Mild | House blend | No sustainability\n\n")

print(wtp_agg %>% mutate(across(c(PartWorth, WTP), ~ round(.x, 3))))


# ---- FIGURE 18: Mean Part-Worth Utilities ----

# I visualised mean part-worths to show the direction and magnitude of consumer

# preferences. Reference levels are shown at zero.
```

```
pw_plot_df <- tribble(
  ~Attribute,      ~Level,               ~PartWorth,
  "Price",         "£3 (ref)",            0,
  "Price",         "£4",                  mean_pw$Price_4,
  "Price",         "£5",                  mean_pw$Price_5,
  "Price",         "£6",                  mean_pw$Price_6,
  "Format",        "Instant (ref)",        0,
  "Format",        "Capsule",             mean_pw$Format_Capsule,
  "Format",        "Ground",              mean_pw$Format_Ground,
  "Format",        "Whole Bean",          mean_pw$Format_Wholebean,
  "Strength",      "Mild (ref)",           0,
  "Strength",      "Medium",              mean_pw$Strength_Medium,
  "Strength",      "Dark",                mean_pw$Strength_Dark,
  "Origin",        "House blend (ref)",     0,
  "Origin",        "100% Arabica",        mean_pw$Origin_Arabica,
  "Origin",        "Single-Origin",       mean_pw$Origin_Single,
  "Sustainability", "No (ref)",             0,
  "Sustainability", "Yes",                  mean_pw$Sustain_Yes
) %>%
  mutate(Attribute = factor(Attribute, levels = c("Price","Format","Strength",
                                          "Origin","Sustainability")),
         Direction = ifelse(PartWorth >= 0, "Positive", "Negative"))


p18 <- ggplot(pw_plot_df, aes(x = reorder(Level, PartWorth), y = PartWorth, fill = Direction)) +
  geom_col(alpha = 0.85, show.legend = FALSE) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey40") +
  geom_text(aes(label = round(PartWorth, 3),
                hjust = ifelse(PartWorth >= 0, -0.1, 1.1)), size = 3.0) +
  coord_flip() +
  facet_wrap(~ Attribute, scales = "free_y", ncol = 2) +
```

```
  scale_fill_manual(values = c("Positive" = "#2E86AB", "Negative" = "#C0392B"))
+
  scale_y_continuous(expand = expansion(mult = c(0.2, 0.2))) +
  labs(title    = "Figure 18: Mean Part-Worth Utilities by Attribute Level",
       subtitle = "Average across 200 respondents; reference levels shown at zero",
       x = "Attribute Level", y = "Part-Worth Utility") +
  theme_assignment()
print(p18)


# ---- FIGURE 19: Willingness to Pay ----
p19 <- ggplot(wtp_agg, aes(x = reorder(Attribute, WTP), y = WTP, fill =
Category)) +
  geom_col(alpha = 0.85) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey40") +
  geom_text(aes(label = paste0("£", round(WTP, 2)),
            hjust = ifelse(WTP >= 0, -0.12, 1.12)), size = 3.2, fontface = "bold") +
  coord_flip() +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = label_dollar(prefix = "£"),
                     expand = expansion(mult = c(0.3, 0.3))) +
  labs(title    = "Figure 19: Willingness to Pay by Attribute Level",
       subtitle = "Aggregate WTP (£) vs reference levels (Instant | Mild | House
blend | No sustainability)",
       x = "Attribute Level", y = "Willingness to Pay (£)", fill = "Attribute") +
  theme_assignment()
print(p19)


# ---------------------------------------------------------------------------
# SECTION 26: STEP 3 — ATTRIBUTE IMPORTANCE
# ---------------------------------------------------------------------------


# I computed attribute importance as the normalised range of part-worths
across
# all levels of each attribute, summing to 100% (Orme, 2006).
```

```
cat("\n=== STEP 3: ATTRIBUTE IMPORTANCE ===\n")

importance_list <- lapply(respondent_ids, function(rid) {
  row <- partworth_df %>% filter(respondent_id == rid)

  range_price    <- max(0, row$Price_4, row$Price_5, row$Price_6, na.rm = TRUE) -
    min(0, row$Price_4, row$Price_5, row$Price_6, na.rm = TRUE)
  range_format   <- max(0, row$Format_Capsule, row$Format_Ground,
                         row$Format_Wholebean, na.rm = TRUE) -
    min(0, row$Format_Capsule, row$Format_Ground,
        row$Format_Wholebean, na.rm = TRUE)
  range_strength <- max(0, row$Strength_Medium, row$Strength_Dark, na.rm = TRUE) -
    min(0, row$Strength_Medium, row$Strength_Dark, na.rm = TRUE)
  range_origin   <- max(0, row$Origin_Arabica, row$Origin_Single, na.rm = TRUE) -
    min(0, row$Origin_Arabica, row$Origin_Single, na.rm = TRUE)
  range_sustain  <- max(0, row$Sustain_Yes, na.rm = TRUE) -
    min(0, row$Sustain_Yes, na.rm = TRUE)

    total <- range_price + range_format + range_strength + range_origin + range_sustain

  data.frame(respondent_id      = rid,
             Imp_Price          = range_price    / total * 100,
             Imp_Format         = range_format   / total * 100,
             Imp_Strength       = range_strength / total * 100,
             Imp_Origin         = range_origin   / total * 100,
             Imp_Sustainability = range_sustain  / total * 100)
})

importance_df <- bind_rows(importance_list)
```

```
importance_summary <- importance_df %>%
  select(-respondent_id) %>%
  summarise(across(everything(), ~ mean(.x, na.rm = TRUE))) %>%
  pivot_longer(everything(), names_to = "Attribute", values_to = "Importance") %>%
  mutate(Attribute = recode(Attribute, "Imp_Price" = "Price", "Imp_Format" = "Format",
                            "Imp_Strength" = "Strength", "Imp_Origin" = "Origin",
                            "Imp_Sustainability" = "Sustainability")) %>%
  arrange(desc(Importance)) %>%
  mutate(Importance = round(Importance, 2))

cat("\n--- Mean Attribute Importance Scores (%) ---\n")
print(importance_summary)
cat("\nSum of importance scores:", round(sum(importance_summary$Importance), 1), "%\n")

# ---- FIGURE 20: Attribute Importance Bar Chart ----
attr_colours <- c("Price" = "#C0392B", "Format" = "#2E86AB", "Strength" = "#3A7D44",
                  "Origin" = "#F4A261", "Sustainability" = "#6A5ACD")

p20 <- ggplot(importance_summary,
              aes(x = reorder(Attribute, Importance), y = Importance, fill = Attribute)) +
  geom_col(alpha = 0.85, show.legend = FALSE) +
  geom_text(aes(label = paste0(round(Importance, 1), "%")),
            hjust = -0.1, size = 4.0, fontface = "bold") +
  coord_flip() +
  scale_fill_manual(values = attr_colours) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.18)),
                     labels = function(x) paste0(x, "%")) +
  labs(title    = "Figure 20: Relative Attribute Importance in Coffee Purchase Decisions",
```

```
      subtitle = "Mean importance scores (%) across 200 respondents; scores sum
to 100%",
       x = "Attribute", y = "Mean Importance (%)") +
  theme_assignment()
print(p20)

importance_sd <- importance_df %>%
  select(-respondent_id) %>%
  summarise(across(everything(), ~ sd(.x, na.rm = TRUE))) %>%
  pivot_longer(everything(), names_to = "Attribute", values_to = "SD") %>%
   mutate(Attribute = recode(Attribute, "Imp_Price" = "Price", "Imp_Format" =
"Format",
                            "Imp_Strength" = "Strength", "Imp_Origin" = "Origin",
                            "Imp_Sustainability" = "Sustainability"))

importance_full <- importance_summary %>%
  left_join(importance_sd, by = "Attribute") %>%
  mutate(SD = round(SD, 2))

cat("\n--- Attribute Importance with Standard Deviation ---\n")
print(importance_full)

write_csv(partworth_df,    "partworth_individual.csv")
write_csv(wtp_agg,         "wtp_aggregate.csv")
write_csv(importance_df,   "importance_individual.csv")
write_csv(importance_full, "importance_summary.csv")




#
=======================================================
=========================
# PART 2 — STEP 4: MARKET SHARE PREDICTION
```

```
# ==========================================================================================

# I used the first-choice rule (maximum utility) to predict market share.
# Each respondent is assumed to choose the product with the highest predicted
# utility. Market share = proportion of respondents choosing each product.

cat("\n=== STEP 4: MARKET SHARE PREDICTION ===\n")

market_profiles <- tribble(
  ~Profile,        ~price, ~format,   ~strength, ~origin,            ~sustainability,
  "Our PB Product",     5, "Capsule", "Dark",    "100% Arabica blend", "Yes",
  "Competitor A",       4, "Capsule", "Medium",  "House blend",        "No",
  "Competitor B",       6, "Capsule", "Dark",    "100% Arabica blend", "No",
  "Competitor C",       5, "Capsule", "Mild",    "Single-origin",      "Yes"
)

cat("\n--- Competitive Set ---\n")
print(market_profiles)

predict_utility <- function(pw_row, price, format, strength, origin, sustain) {
  price_pw    <- case_when(price == 3 ~ 0, price == 4 ~ pw_row$Price_4,
                   price == 5 ~ pw_row$Price_5, price == 6 ~ pw_row$Price_6)
  format_pw    <- case_when(format == "Instant" ~ 0, format == "Capsule" ~ pw_row$Format_Capsule,
                   format == "Ground" ~ pw_row$Format_Ground,
                   format == "Whole bean" ~ pw_row$Format_Wholebean)
  strength_pw <- case_when(strength == "Mild" ~ 0, strength == "Medium" ~ pw_row$Strength_Medium,
                   strength == "Dark" ~ pw_row$Strength_Dark)
  origin_pw   <- case_when(origin == "House blend" ~ 0,
                   origin == "100% Arabica blend" ~ pw_row$Origin_Arabica,
                   origin == "Single-origin" ~ pw_row$Origin_Single)
```

```
  sustain_pw  <- ifelse(sustain == "Yes", pw_row$Sustain_Yes, 0)
    pw_row$Intercept + price_pw + format_pw + strength_pw + origin_pw + sustain_pw
}

utility_matrix <- lapply(respondent_ids, function(rid) {
  pw_row <- partworth_df %>% filter(respondent_id == rid)
  utils  <- sapply(1:nrow(market_profiles), function(i) {
    predict_utility(pw_row, price = market_profiles$price[i],
                    format = market_profiles$format[i],
                    strength = market_profiles$strength[i],
                    origin = market_profiles$origin[i],
                    sustain = market_profiles$sustainability[i])
  })
  names(utils) <- market_profiles$Profile
  as.data.frame(t(utils)) %>% mutate(respondent_id = rid)
}) %>% bind_rows()

cat("\n--- First 6 respondents' predicted utilities ---\n")
print(head(utility_matrix))

utility_matrix <- utility_matrix %>%
       mutate(Choice = apply(select(., -respondent_id), 1, function(x) names(which.max(x))))

market_share <- utility_matrix %>%
  count(Choice) %>%
  mutate(Market_Share_Pct = round(n / sum(n) * 100, 1),
         Choice = factor(Choice, levels = market_profiles$Profile)) %>%
  arrange(Choice)

cat("\n--- Market Share Prediction (First-Choice Rule) ---\n")
print(market_share)
```

```
# ---- FIGURE 21: Market Share ----
share_colours <- c("Our PB Product" = "#2E86AB", "Competitor A" = "#E07B39",
                   "Competitor B" = "#C0392B",   "Competitor C" = "#3A7D44")

p21 <- ggplot(market_share,
              aes(x = reorder(Choice, -Market_Share_Pct), y = Market_Share_Pct, fill =
Choice)) +
  geom_col(alpha = 0.85, show.legend = FALSE) +
  geom_text(aes(label = paste0(Market_Share_Pct, "%\n(n=", n, ")")),
            vjust = -0.3, size = 4.0, fontface = "bold", lineheight = 1.1) +
  scale_fill_manual(values = share_colours) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.18)),
                     labels = function(x) paste0(x, "%")) +
  labs(title    = "Figure 21: Predicted Market Share — First-Choice Rule",
              subtitle = "Capsule, £5, Dark, 100% Arabica, Sustainable vs. three
competitors",
       x = "Product", y = "Predicted Market Share (%)") +
  theme_assignment()
print(p21)

utility_means <- utility_matrix %>%
  select(-respondent_id, -Choice) %>%
  summarise(across(everything(), ~ round(mean(.x, na.rm = TRUE), 3))) %>%
  pivot_longer(everything(), names_to = "Profile", values_to = "Mean_Utility") %>
%
  arrange(desc(Mean_Utility))

cat("\n--- Mean Predicted Utility per Product ---\n")
print(utility_means)

write_csv(market_share,   "market_share_predictions.csv")
write_csv(utility_matrix, "utility_matrix.csv")
```

```
#
==============================================================================
# PART 2 — STEP 5: PCA — MARKET POSITIONING
#
==============================================================================

# I performed PCA on the product attributes dataset to identify the key
# dimensions of consumer perception and position our PB product on a
# perceptual map relative to 24 existing market competitors.

cat("\n=== STEP 5: PCA — MARKET POSITIONING ===\n")

pca_vars <- c("price_100g", "strength_level", "convenience", "authenticity",
              "premium", "perceived_sustainability", "taste_quality")

pca_data <- prod_attr_clean %>% select(product_name, brand, all_of(pca_vars))

cat("PCA input:", nrow(pca_data), "products x", length(pca_vars), "variables\n")

# I standardised all variables before PCA — essential because they are on
# different scales (e.g. price_100g in £ vs perceptual ratings 1–7).
pca_scaled <- pca_data %>% select(all_of(pca_vars)) %>% scale()
rownames(pca_scaled) <- pca_data$product_name

pca_result <- prcomp(pca_scaled, center = FALSE, scale. = FALSE)

cat("\n--- Singular Values ---\n")
cat("(Square roots of eigenvalues of the covariance matrix)\n")
print(round(pca_result$sdev, 4))

pve     <- pca_result$sdev^2 / sum(pca_result$sdev^2)
```

```
cum_pve <- cumsum(pve)

pve_df <- tibble(
  PC          = paste0("PC", 1:length(pve)),
  SingularVal = round(pca_result$sdev, 4),
  Eigenvalue  = round(pca_result$sdev^2, 4),
  PVE         = round(pve * 100, 2),
  Cum_PVE     = round(cum_pve * 100, 2)
)

cat("\n--- PVE Table ---\n")
print(pve_df)
cat(sprintf("\nPC1 + PC2 explain %.1f%% of total variance\n", pve_df$Cum_PVE[2]))

loadings_df <- as.data.frame(pca_result$rotation[, 1:4]) %>%
  rownames_to_column("Variable") %>%
  mutate(Variable = recode(Variable,
                          "price_100g" = "Price/100g", "strength_level" = "Strength",
                         "convenience" = "Convenience", "authenticity" = "Authenticity",
                                "premium" = "Premium", "perceived_sustainability" = "Sustainability",
                          "taste_quality" = "Taste Quality"))

cat("\n--- Loading Matrix (PC1 to PC4) ---\n")
print(loadings_df %>% mutate(across(where(is.numeric), ~ round(.x, 4))))

# ---- FIGURE 22: Scree Plot ----
p22 <- ggplot(pve_df, aes(x = PC, y = PVE, group = 1)) +
  geom_line(colour = "#2E86AB", linewidth = 1.3) +
  geom_point(colour = "#2E86AB", size = 4) +
  geom_col(aes(y = PVE), fill = "#2E86AB", alpha = 0.15) +
  geom_text(aes(label = paste0(PVE, "%")), vjust = -0.6, size = 3.2, fontface = "bold") +
```

```
  scale_y_continuous(labels = function(x) paste0(x, "%"),
                     expand = expansion(mult = c(0, 0.15))) +
  labs(title    = "Figure 22: PCA Scree Plot — Proportion of Variance Explained",
       subtitle = "Each bar/point shows the % variance captured by each principal component",
       x = "Principal Component", y = "Proportion of Variance Explained (%)") +
  theme_assignment()
print(p22)


# ---- FIGURE 23: Loading Plot ----
# I visualised PC1 and PC2 loadings to interpret what each component represents.
# PC1 = Mainstream (Convenience/Strength/Price) vs Artisan (Authenticity/Sustainability).
# PC2 = Taste Quality / Sensory dimension.


loading_plot_df <- loadings_df %>% select(Variable, PC1, PC2)


p23 <- ggplot(loading_plot_df, aes(x = PC1, y = PC2, label = Variable)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_segment(aes(x = 0, y = 0, xend = PC1, yend = PC2),
               arrow = arrow(length = unit(0.25, "cm")),
               colour = "#2E86AB", linewidth = 1.0) +
  geom_text(nudge_x = 0.02, nudge_y = 0.03, size = 3.5, fontface = "bold", colour = "grey20") +
  coord_cartesian(xlim = c(-0.65, 0.65), ylim = c(-0.65, 0.65)) +
  labs(title    = "Figure 23: PCA Loading Plot — PC1 vs PC2",
             subtitle = "Arrows show variable contributions; direction indicates association",
       x = paste0("PC1 (", pve_df$PVE[1], "% variance)"),
       y = paste0("PC2 (", pve_df$PVE[2], "% variance)")) +
  theme_assignment()
print(p23)
```

```
# ---- FIGURE 24: Biplot + Perceptual Map ----
# I constructed a biplot overlaying product scores with loading arrows.
# Our PB product is projected onto the space using assumed perceptual values:
# £6.50/100g, strength 7 (dark), convenience 6.5, authenticity 4.0,
# premium 5.0, perceived sustainability 5.5, taste quality 5.5.

scores_df <- as.data.frame(pca_result$x[, 1:2]) %>%
  rownames_to_column("product_name") %>%
  left_join(pca_data %>% select(product_name, brand), by = "product_name")

scale_factor <- max(abs(scores_df[, c("PC1","PC2")])) /
  max(abs(pca_result$rotation[, 1:2])) * 0.6

arrow_df <- loading_plot_df %>%
  mutate(PC1_scaled = PC1 * scale_factor, PC2_scaled = PC2 * scale_factor)

pb_product <- data.frame(price_100g = 6.50, strength_level = 7, convenience =
6.5,
                         authenticity = 4.0, premium = 5.0,
                         perceived_sustainability = 5.5, taste_quality = 5.5)

pca_means <- attr(pca_scaled, "scaled:center")
pca_sds   <- attr(pca_scaled, "scaled:scale")
pb_scaled <- (pb_product - pca_means) / pca_sds
pb_scores <- as.matrix(pb_scaled) %*% pca_result$rotation[, 1:2]

pb_df <- data.frame(product_name = "Our PB Product",
                    PC1 = pb_scores[1, 1], PC2 = pb_scores[1, 2], brand = "PB")

all_scores <- bind_rows(scores_df, pb_df) %>%
  mutate(IsOurs = product_name == "Our PB Product")
```

```
brand_colours <- c("Costa" = "#3A7D44", "KENCO" = "#E07B39", "Lavazza" = "#6A5ACD",
                   "Nescafe" = "#F4A261", "Starbucks" = "#2E86AB", "Taylors" = "#1A5276",
              "Waitrose" = "#8E44AD", "PB" = "#C0392B")


p24 <- ggplot() +
  geom_segment(data = arrow_df,
               aes(x = 0, y = 0, xend = PC1_scaled, yend = PC2_scaled),
               arrow = arrow(length = unit(0.2, "cm")), colour = "grey50", linewidth = 0.8) +
  geom_text(data = arrow_df,
            aes(x = PC1_scaled * 1.12, y = PC2_scaled * 1.12, label = Variable),
            size = 2.8, colour = "grey30", fontface = "italic") +
  geom_point(data = all_scores,
             aes(x = PC1, y = PC2, colour = brand, size = IsOurs, shape = IsOurs)) +
  geom_text(data = all_scores,
            aes(x = PC1, y = PC2 + 0.12, label = product_name, colour = brand),
            size = 2.5, show.legend = FALSE) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey70") +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "grey70") +
  scale_colour_manual(values = brand_colours) +
  scale_size_manual(values  = c("FALSE" = 2.5, "TRUE" = 5.0)) +
  scale_shape_manual(values = c("FALSE" = 16,  "TRUE" = 18)) +
  labs(title    = "Figure 24: Perceptual Map — PCA Biplot (PC1 vs PC2)",
       subtitle = "Product scores + attribute loadings; red diamond = our proposed PB product",
       x = paste0("PC1 (", pve_df$PVE[1], "% variance)"),
       y = paste0("PC2 (", pve_df$PVE[2], "% variance)"), colour = "Brand") +
  guides(size = "none", shape = "none") +
  theme_assignment()
print(p24)


write_csv(pve_df,      "pca_pve.csv")
```

```
write_csv(loadings_df, "pca_loadings.csv")
write_csv(scores_df,   "pca_scores.csv")
```